

Larynx - Notes

Comprehensive anatomical, physiological, and clinical coverage of the larynx, including cartilages, joints, membranes, muscles, cavity subdivisions, vasculature, innervation, phonation, and key clinical correlations.

Chapter Snapshot

The larynx is the expanded upper part of the lower respiratory tract, serving as the organ of phonation and a protective sphincter for the trachea. Located in the anterior midline of the neck (C3–C6), it is a complex mechanism of cartilages, ligaments, and muscles that regulates airflow and produces voice.

Key Structural Components:

- **Skeleton:** Nine cartilages (3 unpaired, 3 paired).
- **Framework:** Thyroid (shield), Cricoid (foundation), Epiglottis (lid).
- **Mobility:** Joints and intrinsic/extrinsic muscles control vocal cord tension and position.
- **Cavity:** Divided into vestibule, ventricle, and infraglottic regions by false and true vocal folds.

Primary Functions:

1. **Protection:** Prevents aspiration of food/fluid into the lower respiratory tract.
2. **Phonation:** Generates sound via vocal cord vibration.
3. **Respiration:** Regulates air entry.
4. **Deglutition:** Assists in swallowing via upward/downward movement.

Exam Focus: The primary function of the larynx is **protection**, not phonation. Phonation is a later evolutionary development. Always prioritize the protective role in clinical reasoning questions.

Why This Chapter Matters

Understanding the larynx is critical for two main reasons:

1. **Clinical Emergencies:** The larynx is a common site for airway obstruction. Knowledge of the narrowest point (glottis) and the mechanism of aspiration is vital for managing choking (Heimlich maneuver) and understanding laryngeal incompetence.

2. **Voice and Nerve Injury:** The nerve supply is complex and vulnerable. Injury to the recurrent laryngeal nerve (e.g., during thyroidectomy) causes hoarseness or airway obstruction. Understanding muscle actions (abduction vs. adduction) allows you to predict clinical signs of paralysis.

Must Remember: The larynx is the only part of the respiratory tract with voluntary control (via motor speech area), allowing for speech and cough reflexes.

High-Yield Identity

Location: Anterior midline of neck, C3–C6 vertebrae.

Sex Differences:

- **Male:** Larger, lower pitched voice, acute thyroid angle (90°), prominent Adam's apple.
- **Female/Child:** Smaller, higher pitched voice, obtuse thyroid angle (120°), less prominent prominence.

Cartilage Types:

- **Hyaline:** Thyroid, Cricoid, basal Arytenoids. (Ossify after age 25; visible on X-ray).
- **Elastic:** Epiglottis, Corniculate, Cuneiform, apical Arytenoids. (Do not ossify; remain flexible).

Key Landmarks:

- **Adam's Apple:** Laryngeal prominence of thyroid cartilage.
- **Foundation Stone:** Cricoid cartilage (complete ring).
- **Safety Muscles:** Posterior cricoarytenoids (only abductors; open airway in emergency).

Core Framework

To master the larynx, visualize it as a **dynamic tube** with three functional layers:

1. The Skeleton (Static Support)

- **Thyroid:** Anterior shield.

- **Cricoid:** Posterior support and anterior arch (C6 level).
- **Epiglottis:** Posterior flap for protection.
- **Arytenoids:** Mobile pyramids that rotate/glides to move cords.

2. The Connective Tissue (Functional Membranes)

Two key membranes define the internal architecture:

- **Quadrangular Membrane:** Upper part. Forms **vestibular folds** (false cords).
- **Cricovocal Membrane (Conus Elasticus):** Lower part. Forms **vocal folds** (true cords).
- **Sinus of Larynx:** The space between these two membranes.

3. The Musculature (Active Control)

Muscles act on the arytenoids and thyroid to change cord length, tension, and position:

- **Abductors:** Posterior cricoarytenoid (opens glottis).
- **Adductors:** Lateral cricoarytenoid, Transverse arytenoid, Thyroarytenoid (closes glottis).
- **Tensors:** Cricothyroid (tenses/lengthens), Vocalis (modulates pitch).

Clinical Focus: The **Rima Glottidis** is the narrowest part of the larynx. Its shape changes with function:

- **Quiet breathing:** Pentagonal.
- **Inspiration:** Diamond-shaped (widest).
- **Phonation:** Linear chink (narrowest).
- **Whispering:** Inverted funnel.

Warning: Do not confuse the **vestibular folds** (false cords) with **vocal folds** (true cords). Only the true cords produce sound. The false cords are for protection and closure.

Detailed Notes

Introduction: Functions and Primary Role

Definition and Core Identity

The larynx is the upper expanded portion of the lower respiratory tract. It is anatomically modified to serve as the organ of phonation (voice box).

Structurally, it acts as a dynamic sphincter at the inlet of the lower respiratory tract. This sphincteric action is critical for regulating airflow and preventing foreign material from entering the trachea and bronchial tree.

Key Point: The larynx is not a static tube; it is a complex muscular-cartilaginous framework designed for rapid, coordinated movement.

The Four Functions

The larynx performs four distinct physiological roles. For exam purposes, these must be listed clearly:

1. **Phonation:** Production of voice/sound.
2. **Respiration:** Providing a patent airway for the passage of air.
3. **Protection:** Guarding the lower respiratory tract from aspiration.
4. **Deglutition:** Assisting in the act of swallowing.

Primary Function: Protection vs. Phonation

A common student error is assuming phonation is the primary function because it is the most obvious clinical feature. The source text explicitly corrects this hierarchy.

Must Remember: The primary (most important) function of the larynx is **Protection** of the lower respiratory tract.

Why Protection is Primary:

- **Evolutionary Logic:** Phonation is a later evolutionary development linked to the motor speech area of the brain. Protection is a fundamental survival mechanism present in earlier evolutionary stages.
- **Physiological Priority:** If the protective mechanism fails, the result is life-threatening aspiration. If phonation fails, communication is impaired, but survival is not immediately threatened.

Clinical Consequence: Laryngeal Incompetence

When the protective sphincter action of the larynx is deranged, **laryngeal incompetence** occurs.

- **Mechanism:** Failure of the laryngeal inlet to close tightly during swallowing or coughing.
- **Result:** Food and fluids are aspirated into the trachea and bronchial tree.
- **Clinical Impact:** This leads to aspiration pneumonia, choking, and chronic respiratory complications.

Role in Deglutition (Swallowing)

The larynx actively participates in swallowing through vertical movement.

- **Upward Movement:** Elevates the larynx to meet the descending epiglottis and soft palate, helping to seal off the nasopharynx and direct food into the esophagus.
- **Downward Movement:** Returns the larynx to its resting position after the swallow is complete.

Exam Answer:

"The larynx serves four functions: phonation, respiration, protection, and deglutition. Its primary evolutionary function is protection of the lower respiratory tract from aspiration. Failure of this protective mechanism results in laryngeal incompetence, leading to aspiration of food and fluids."

Location, Extent, and Sex Differences

Anatomical Position and Vertebral Relations

The larynx is situated in the **anterior midline** of the upper neck. It lies directly in front of the **laryngopharynx**.

Vertical Extent:

- **Superior limit:** Root of the tongue.
- **Inferior limit:** Trachea.
- **Vertebral level:** It lies anterior to the **C3, C4, C5, and C6** cervical vertebrae.

Key Point: The larynx occupies the C3–C6 vertebral levels. This is a fixed anatomical landmark for neck dissections and radiological identification.

Size Dimensions and Sexual Dimorphism

The larynx exhibits significant sexual dimorphism. While pre-pubertal larynxes are similar in size between sexes, male larynxes grow rapidly at puberty, resulting in a larger adult structure compared to females.

Dimension	Male (Average)	Female (Average)
Vertical	44 mm	36 mm
Transverse	43 mm	41 mm
Anteroposterior	36 mm	26 mm

Exam Focus: Note that the female larynx is shorter vertically but nearly equal in transverse width. The most significant difference is in the anteroposterior depth and vertical height, reflecting the larger male cartilaginous framework.

Pubertal Changes and Voice Pitch

Puberty drives the structural divergence between male and female larynxes, directly affecting voice characteristics.

- **Male Larynx:**
- Undergoes rapid growth during puberty.

- The **thyroid cartilage angle** becomes more acute (approx. **90°**).
- This acute angle creates the prominent **laryngeal prominence** (Adam's apple).
- The larger size and longer vocal cords result in a **louder, lower-pitched** voice.
- **Female Larynx:**
 - Size remains similar to the pre-pubertal (child) larynx.
 - The **thyroid cartilage angle** is obtuse (approx. **120°**).
 - The lack of significant growth maintains shorter vocal cords, resulting in a **higher-pitched** voice.

Clinical Focus: The "Adam's apple" is not a separate structure but the prominent anterior projection of the thyroid cartilage laminae meeting at an acute angle. This prominence is absent or subtle in females due to the wider, obtuse angle.

Summary Checklist for Exam Revision

- **Location:** Anterior midline neck, C3–C6.
- **Relations:** Anterior to laryngopharynx.
- **Sex Difference:** Male larynx larger; female larynx smaller (child-like size).
- **Voice Correlation:** Male = Low pitch (large larynx, acute angle); Female = High pitch (small larynx, obtuse angle).
- **Must Remember:** The larynx is higher in children and females compared to adult males.

The Skeleton: Unpaired Cartilages (Thyroid, Cricoid, Epiglottis)

The laryngeal skeleton consists of nine cartilages. Three are unpaired and form the anterior and posterior framework: the thyroid, cricoid, and epiglottis. The remaining six are paired. Understanding these three is essential for grasping laryngeal mechanics, phonation, and airway protection.

Thyroid Cartilage: The Shield

The thyroid cartilage is the largest laryngeal cartilage. It sits anteriorly and forms the bulk of the laryngeal prominence.

Structure and Shape

- It consists of two quadrilateral laminae that meet in the midline.
- The laminae fuse anteriorly to form the **thyroid angle**.
- **Sex Differences:** This angle is a key sex dimorphism.
- **Males:** The angle is acute (~90°), creating a prominent projection called the **Adam's apple** (laryngeal prominence).
- **Females:** The angle is wider (~120°), resulting in a smoother neck contour.
- **Horns (Cornu):** The posterior borders of the laminae extend superiorly and inferiorly as horns.
- **Superior cornu:** Articulates with the hyoid bone via the thyrohyoid membrane.
- **Inferior cornu:** Articulates with the cricoid cartilage to form the cricothyroid joint.

Attachments (Exam Focus) The thyroid cartilage provides attachment sites for multiple muscles and ligaments. Memorize these by surface:

- **Posterior Surface (Median Plane):** From above downward, it attaches to:

1. Thyroepiglottic ligament.
2. Vestibular ligaments (false cords).
3. Vocal ligaments (true cords).

- **Posterior Border of Laminae:** Provides conjoined insertion for three pharyngeal muscles:

1. Palatopharyngeus.
2. Salpingopharyngeus.
3. Stylopharyngeus.

- **Outer Surface (Oblique Line):** From above downward, it attaches to:

1. Thyrohyoid muscle.
2. Sternothyroid muscle.
3. Inferior constrictor (thyropharyngeus part).

Clinical Focus: Fractures of the thyroid cartilage are common in anterior neck trauma. The superior horns are palpable landmarks for identifying the level of the hyoid bone and the cricoid cartilage.

Cricoid Cartilage: The Foundation Stone

The cricoid cartilage is the only complete ring of cartilage in the respiratory tract. It sits inferior to the thyroid cartilage at the level of the **C6 vertebra**.

Structure

- It is shaped like a signet ring.
- **Anterior Arch:** Narrow and short.
- **Posterior Lamina:** Broad and tall.
- **Median Ridge:** A vertical ridge on the posterior surface of the lamina divides the posterior aspect into two depressions. This ridge serves as the origin for the posterior cricoarytenoid muscles.

Significance

- **Foundation Stone:** It is called the "foundation stone" of the larynx because it is the only cartilage that forms a complete ring. This structural integrity prevents the larynx from collapsing during negative inspiratory pressures.
- **Articulation:** The lateral parts of the upper border of the lamina articulate with the bases of the arytenoid cartilages, forming the cricoarytenoid joints.

Must Remember: The cricoid cartilage is the only complete ring in the airway. This makes it a critical landmark in airway management; damage here can lead to catastrophic airway collapse.

Epiglottis: The Lid

The epiglottis is a leaf-shaped plate of elastic cartilage that acts as a valve for the laryngeal inlet.

Shape and Position

- It is located behind the tongue and hyoid bone.
- **Broad Upper End:** Free and curved. It forms the upper boundary of the laryngeal inlet.
- **Narrow Lower End (Stalk):** Pointed and attached to the posterior surface of the thyroid angle by the **thyroepiglottic ligament**.

Attachments and Relations

- **Anterior Surface:** Connected to the base of the tongue by median and lateral glossoepiglottic folds.
- **Valleculae:** The depressions on either side of the median glossoepiglottic fold are called valleculae. These are clinically significant as they are a common site for foreign body impaction in children.
- **Posterior Surface:** Presents a tubercle in its lower part.

Function

- During deglutition (swallowing), the epiglottis folds down to cover the laryngeal inlet, preventing food and liquid from entering the trachea.
- In humans, the epiglottis is relatively small. In macrosomatic animals (e.g., cattle), it is elongated and extends beyond the soft palate into the nasopharynx to prevent water inhalation while drinking.

Key Point: The epiglottis is made of **elastic cartilage**, allowing it to be flexible and spring back to its upright position after swallowing.

Cartilage Types and Ossification

The composition of these cartilages changes with age and varies by structure.

- **Hyaline Cartilage:**
 - Found in: Thyroid, Cricoid, and the basal parts of the Arytenoids.
 - **Clinical Significance:** Hyaline cartilage tends to **ossify** (turn to bone) after age 25. This ossification can be seen on radiographs (X-rays) and may complicate surgical procedures or mask pathology.
- **Elastic Cartilage:**
 - Found in: Epiglottis, Corniculate, Cuneiform, and the apices of the Arytenoids.
 - **Clinical Significance:** Elastic cartilage does **not** ossify. It remains flexible throughout life, which is essential for the dynamic function of the epiglottis and the small cartilages.

Exam Answer: The thyroid and cricoid cartilages are composed of hyaline cartilage, which ossifies with age. The epiglottis is composed of elastic cartilage, which remains flexible. This difference is critical for understanding laryngeal rigidity versus flexibility.

Summary Table: Unpaired Cartilages

Feature	Thyroid Cartilage	Cricoid Cartilage	Epiglottis
Shape	Box-like with anterior angle	Signet ring	Leaf-shaped
Type	Hyaline	Hyaline	Elastic
Key Landmark	Adam's apple (males)	Foundation stone	Laryngeal inlet lid
Ossification	Yes (after 25 yrs)	Yes (after 25 yrs)	No
Primary Function	Framework, protection	Structural integrity, joint base	Airway protection (swallowing)
Attachments	Hyoid (sup), Cricoid (inf)	Trachea (inf), Arytenoids (sup)	Thyroid angle (inf), Tongue (ant)

Warning: Do not confuse the cricoid cartilage with the thyroid cartilage. The cricoid is inferior and complete; the thyroid is superior and open posteriorly. This distinction is vital for intubation and airway surgery.

The Skeleton: Paired Cartilages (Arytenoid, Corniculate, Cuneiform)

The larynx contains three pairs of small cartilages that play critical roles in mobility, protection, and the structural integrity of the posterior larynx. Unlike the large thyroid and cricoid, these cartilages are smaller, more mobile, and often overlooked in superficial reviews. However, they are essential for understanding vocal cord movement and laryngeal closure.

Arytenoid Cartilages: The Mobile Handles

The arytenoid cartilages are the most functionally significant of the paired cartilages. They act as the mechanical "handles" for the vocal cords. Understanding their shape and processes is the key to understanding how the larynx opens and closes.

Shape and Orientation

- **Pyramidal Structure:** Each arytenoid is a three-sided pyramid.
- **Base:** The broad base rests on the superior border of the cricoid lamina. This articulation forms the **cricoarytenoid joint**, allowing rotation and gliding.
- **Apex:** The apex is curved posteromedially and serves as the attachment point for the corniculate cartilage.
- **Surfaces:**
 - **Posterior surface:** Smooth, convex, and provides attachment for the transverse and oblique arytenoid muscles.
 - **Medial surface:** Faces the opposite arytenoid; separated by the interarytenoid space.
 - **Anterolateral surface:** Concave, providing attachment for the lateral cricoarytenoid and vocalis muscles.

Key Processes: The Exam Trap The base of the arytenoid extends into two distinct processes. Confusing these two is a common error.

- **Vocal Process:**

- **Direction:** Projects forward (anteriorly).
- **Function:** Provides attachment for the **vocal ligament** (true vocal cord).
- **Clinical Relevance:** When the posterior cricoarytenoid muscle pulls the muscular process backward, the entire arytenoid rotates, causing the vocal process to swing outward, abducting the vocal cords.

- **Muscular Process:**

- **Direction:** Projects laterally and backward.
- **Function:** Provides attachment for the **posterior cricoarytenoid** (abductor) and **lateral cricoarytenoid** (adductor) muscles.
- **Clinical Relevance:** This is the lever arm for vocal cord movement.

Must Remember: The arytenoid cartilages are the only laryngeal cartilages that move significantly during phonation. Their rotation around a vertical axis is the mechanism for abducting and adducting the vocal cords.

Corniculate Cartilages (of Santorini)

The corniculate cartilages are small, conical nodules that articulate with the apices of the arytenoid cartilages.

- **Location:** They lie within the posterior part of the **aryepiglottic folds**.
- **Structure:** Composed of elastic cartilage.
- **Function:** They reinforce the aryepiglottic folds, helping to close the laryngeal inlet during swallowing.
- **Joint:** They articulate with the arytenoid apex via the **arytenocorniculate joint**. This is a synovial joint but has **no functional significance** in movement.

Cuneiform Cartilages (of Wrisberg)

The cuneiform cartilages are tiny, rod-shaped structures.

- **Location:** They lie in the posterior part of the **aryepiglottic folds**, just **above** the corniculate cartilages.
- **Structure:** Composed of elastic cartilage.
- **Function:** Like the corniculate cartilages, they stiffen the aryepiglottic folds to aid in laryngeal closure during deglutition.
- **Visibility:** They are often palpable as small elevations in the aryepiglottic folds during laryngoscopy.

Key Point: Do not confuse the cuneiform and corniculate cartilages. The **corniculate** articulates with the **arytenoid** (below). The **cuneiform** lies **above** the corniculate, free in the fold.

Cartilage Types: Hyaline vs. Elastic

The laryngeal skeleton is not uniform in composition. The type of cartilage determines ossification patterns and radiographic visibility.

Cartilage	Type	Ossification	Radiographic Visibility
Thyroid	Hyaline	Yes (>25 years)	Visible
Cricoid	Hyaline	Yes (>25 years)	Visible
Arytenoid (Base)	Hyaline	Yes (>25 years)	Visible
Arytenoid (Apex)	Elastic	No	Not visible
Epiglottis	Elastic	No	Not visible
Corniculate	Elastic	No	Not visible
Cuneiform	Elastic	No	Not visible

Exam Focus: Hyaline cartilage ossifies with age (typically after 25 years), which can be seen on X-rays as calcification of the thyroid and cricoid cartilages. Elastic cartilage does not ossify. This distinction is critical for interpreting radiographs and understanding age-related changes in the larynx.

Summary Checklist for Revision

- **Arytenoid:** Pyramidal, base on cricoid, apex with corniculate. Two processes: **Vocal** (forward, for vocal ligament) and **Muscular** (lateral/back, for muscles).
- **Corniculate:** Small, conical, elastic, articulates with arytenoid apex.
- **Cuneiform:** Tiny, rod-shaped, elastic, lies above corniculate in aryepiglottic fold.
- **Cartilage Types:** Thyroid/Cricoid/Arytenoid base = Hyaline (ossifies). Epiglottis/Rest of Arytenoid/Corniculate/Cuneiform = Elastic (no ossification).
- **Function:** Arytenoids are the mobile handles for vocal cord abduction/adduction. Corniculate/Cuneiform stiffen the aryepiglottic folds for laryngeal closure.

Exam Answer: The paired cartilages of the larynx are the arytenoid, corniculate, and cuneiform. The arytenoids are pyramidal hyaline cartilages (base) that articulate with the cricoid lamina and serve as the mobile handles for vocal cord movement via their vocal and muscular processes. The corniculate and cuneiform cartilages are small elastic cartilages located in the aryepiglottic folds, reinforcing the laryngeal inlet during swallowing.

Joints of the Larynx

The laryngeal skeleton is not a rigid block; its cartilages are connected by three pairs of synovial joints. These joints provide the mechanical axes necessary for the intrinsic muscles to adjust the vocal cords for phonation and respiration. Understanding the specific movement permitted by each joint is essential for explaining how the larynx functions as a dynamic valve.

Cricothyroid Joint

This is a plane synovial joint formed between the inferior cornu (horn) of the thyroid cartilage and the lateral aspect of the cricoid cartilage. It is a critical anatomical landmark because the recurrent laryngeal nerve enters the larynx in immediate proximity to this joint.

The joint permits two distinct types of movement that work together to tension the vocal cords:

- **Rotatory Movement:** The cricoid cartilage rotates on the inferior cornua of the thyroid cartilage around a transverse axis. This axis passes transversely through both cricothyroid joints.
- **Gliding Movement:** The cricoid cartilage glides in various directions relative to the thyroid cornua.

Key Point: The combined rotation and gliding at the cricothyroid joint allow the thyroid cartilage to tilt downwards and forwards relative to the cricoid. This action lengthens and tenses the vocal ligaments, raising the pitch of the voice.

Cricoarytenoid Joint

This is a tiny plane synovial joint located between the base of the arytenoid cartilage and the lateral part of the upper border of the lamina of the cricoid cartilage. It is the primary joint responsible for the opening and closing of the glottis.

The joint permits two types of movement:

- **Rotatory Movement:** The arytenoid cartilage moves around a vertical axis. This rotation causes the vocal process to move medially or laterally, resulting in the adduction or abduction of the vocal cords.
- **Gliding Movement:** One arytenoid cartilage glides towards or away from the other. This sliding motion assists in closing or opening the posterior part of the glottis.

Exam Focus: The cricoarytenoid joint is the mechanical pivot for the posterior cricoarytenoid muscle (abduction) and the lateral cricoarytenoid muscle (adduction). Without this joint, the vocal cords could not open for breathing or close for phonation.

Arytenocorniculate Joint

This is a synovial joint formed between the apex of the arytenoid cartilage and the corniculate cartilage.

- **Function:** It is of no functional significance in laryngeal mechanics.
- **Structure:** It allows slight mobility of the corniculate cartilage within the aryepiglottic fold but does not contribute to phonation or respiration.

Warning: Do not confuse this joint with the functional cricoarytenoid joint. The arytenocorniculate joint is often tested as a distractor in anatomy exams because it lacks clinical or functional importance compared to the other two joints.

Comparative Summary of Laryngeal Joints

To ensure clarity and aid revision, the three joints are summarized below by their articulation, type, and functional significance.

Joint	Articulation	Type	Movement Permitted	Functional Significance
Cricothyroid	Inferior cornu of thyroid cartilage ↔ Cricoid cartilage	Plane Synovial	Rotatory (transverse axis) and Gliding	Tenses vocal cords; raises pitch.
Cricoarytenoid	Base of arytenoid ↔ Upper border of cricoid lamina	Plane Synovial	Rotatory (vertical axis) and Gliding	Abducts/adducts vocal cords; opens/closes glottis.
Arytenocorniculate	Apex of arytenoid ↔ Corniculate cartilage	Synovial	Negligible	No functional significance.

Exam Answer: When asked to describe the joints of the larynx, state that there are three pairs of synovial joints. Identify the cricothyroid and cricoarytenoid joints as functionally vital for phonation and respiration, specifying their axes of rotation. Note the arytenocorniculate joint as non-functional. This structure demonstrates a clear understanding of laryngeal biomechanics.

Ligaments and Membranes: Extrinsic vs Intrinsic

The laryngeal skeleton is stabilized by fibrous membranes and ligaments. These structures are classified into two distinct groups based on their relationship to the laryngeal cavity: **extrinsic** (external attachments) and **intrinsic** (internal framework).

Exam Focus: Distinguish between the "outer tube" (extrinsic) and the "inner fibroelastic tube" (intrinsic).

Classification Overview

The ligaments and membranes are not random; they form a structural scaffold.

- **Extrinsic structures:** Connect the larynx to adjacent neck structures (hyoid bone, trachea). They suspend and stabilize the larynx as a whole unit.
- **Intrinsic structures:** Form the internal wall of the larynx. They define the laryngeal inlet, vestibule, and ventricles. They contain the functional ligaments (vocal and vestibular).

Key Point: The intrinsic membranes are interrupted by the sinus of the larynx (ventricle), creating two distinct zones: the upper quadrangular membrane and the lower cricovocal membrane.

Extrinsic Ligaments and Membranes

These structures lie outside the fibroelastic tube of the larynx. They are primarily suspensory or connective.

1. Thyrohyoid Membrane and Ligaments

- Extends from the upper border of the thyroid cartilage to the upper border of the hyoid bone.
- Thickens medially and laterally to form the median and lateral thyrohyoid ligaments.
- **Clinical Note:** The lateral thyrohyoid ligament contains a small nodule of elastic cartilage called the *cartilago triticea*.
- **Transit:** Pierced by the internal laryngeal nerve and superior laryngeal vessels.

1. Cricotracheal Ligament

- Connects the cricoid cartilage to the first tracheal ring.
- Ensures continuity between the larynx and the trachea.

1. Thyroepiglottic Ligament

- Attaches the lower narrow end (stalk) of the epiglottis to the posterior surface of the thyroid angle.

1. Hyoepiglottic Ligament

- Connects the posterior aspect of the hyoid bone to the anterior surface of the upper end of the epiglottis.

1. Cricothyroid Ligament

- Connects the lower border of the thyroid cartilage to the cricoid cartilage in the midline.

Intrinsic Ligaments and Membranes

The intrinsic structures form a broad sheet of fibroelastic tissue that constitutes the inner tube of the laryngeal cavity. This tube is divided into two parts by the sinus of the larynx (the space between the true and false vocal cords).

• 1. Quadrangular Membrane (Above the Sinus)

- **Extent:** Extends from the sides of the epiglottis to the arytenoid cartilages.
- **Lower Edge:** Free and thickened to form the **Vestibular Ligament**.
- **Attachment:** The vestibular ligament extends from the posterior surface of the thyroid cartilage (above the cricothyroid membrane) to the lateral surface of the arytenoid cartilage (in front of the muscular process).
- **Function:** Supports the vestibular fold (false vocal cord).

• 2. Cricovocal Membrane / Conus Elasticus (Below the Sinus)

- **Extent:** Extends upwards and medially from the upper border of the arch of the cricoid cartilage.
- **Upper Edge:** Free and thickened to form the **Vocal Ligament**.
- **Attachment:** The vocal ligament extends from the posterior surface of the thyroid cartilage to the vocal process of the arytenoid cartilage.
- **Function:** Supports the vocal fold (true vocal cord).

Must Remember: The cricovocal membrane is also known as the **Conus Elasticus**. It is the foundational support for the true vocal cords.

Functional Ligaments: Vestibular vs. Vocal

The free edges of the intrinsic membranes form two critical ligaments. These are not separate structures but thickened edges of the membranes.

Feature	Vestibular Ligament	Vocal Ligament
Parent Membrane	Lower edge of Quadrangular Membrane	Upper edge of Cricovocal Membrane (Conus Elasticus)
Location	Superior (False Cord)	Inferior (True Cord)
Attachments	Thyroid cartilage to Arytenoid (lateral surface)	Thyroid cartilage to Arytenoid (vocal process)
Composition	Fibrous tissue	Yellow elastic tissue
Clinical Significance	Supports false vocal fold (respiration/protection)	Supports true vocal fold (phonation)

Exam Answer: The vestibular ligament is the thickened lower edge of the quadrangular membrane, while the vocal ligament is the thickened upper edge of the cricovocal membrane.

Summary Table: Extrinsic vs. Intrinsic

Category	Structure	Key Relations/Notes
Extrinsic	Thyrohyoid Membrane	Pierced by internal laryngeal nerve/vessels. Contains cartilago triticea laterally.
Extrinsic	Cricotracheal Ligament	Connects cricoid to 1st tracheal ring.
Extrinsic	Thyroepiglottic Ligament	Epiglottis to thyroid angle.
Extrinsic	Hyoepiglottic Ligament	Hyoid to epiglottis.
Extrinsic	Cricothyroid Ligament	Thyroid to cricoid (midline).
Intrinsic	Quadrangular Membrane	Forms vestibular ligament (false cord support).
Intrinsic	Cricovocal Membrane	Forms vocal ligament (true cord support). Also called Conus Elasticus.

Warning: Do not confuse the *cricothyroid ligament* (extrinsic, midline connection) with the *cricothyroid membrane* (lateral joint space). The extrinsic ligament is midline; the joint is lateral.

Exam Focus: In exams, if asked for the "foundation of the vocal cord," the answer is the **vocal ligament**, which is part of the **cricovocal membrane** (Conus Elasticus). If asked for the structure supporting the false cord, it is the **vestibular ligament** from the **quadrangular membrane**.

Muscles: Extrinsic and Functional Groups

The laryngeal muscles are divided into two distinct functional groups: extrinsic muscles that move the larynx as a whole unit, and intrinsic muscles that move the cartilages relative to each other to control phonation and airway protection.

Key Point: Extrinsic muscles act on the larynx's position; intrinsic muscles act on the larynx's function (sound and airflow).

Extrinsic Muscles: Positional Control

The extrinsic muscles attach the larynx to surrounding structures (hyoid bone, sternum, clavicle, pharynx). Their primary role is elevating or depressing the larynx during swallowing and speech.

- **Elevators (4 muscles):**

- Palatopharyngeus
- Salpingopharyngeus
- Stylopharyngeus
- Thyrohyoid

- **Depressor (1 muscle):**

- Sternothyroid

Exam Focus: The Thyrohyoid is the only extrinsic muscle that elevates the larynx by pulling the thyroid cartilage upward toward the hyoid bone. The Sternothyroid is the only extrinsic muscle that depresses the larynx.

Note: Detailed anatomy of the pharyngeal muscles (Palatopharyngeus, Salpingopharyngeus, Stylopharyngeus) is covered in the Pharynx chapter. Here, focus on their collective role in laryngeal elevation.

Intrinsic Muscles: Functional Grouping

The intrinsic muscles connect laryngeal cartilages to each other. They do not move the larynx in the neck; they manipulate the vocal cords and laryngeal inlet.

Must Remember: All intrinsic muscles are paired, **except** the Transverse Arytenoid, which is unpaired.

To simplify revision, classify intrinsic muscles by their primary action:

1. **Open/Close the Laryngeal Inlet**
2. **Abduct/Adduct the Vocal Cords**
3. **Tense/Relax the Vocal Cords**

Group 1: Inlet Control (Obstruction/Opening)

These muscles manipulate the epiglottis and arytenoids to protect the airway during swallowing.

Muscle	Action	Clinical/Exam Note
Oblique Arytenoid	Closes the laryngeal inlet	Works with aryepiglotticus
Aryepiglotticus	Closes the laryngeal inlet	Forms the aryepiglottic fold
Thyroepiglotticus	Opens the laryngeal inlet	Upper fibers of Thyroarytenoid

Teaching Point: The Oblique Arytenoid and Aryepiglotticus act as a sphincter to close the inlet when swallowing. The Thyroepiglotticus (often considered part of the Thyroarytenoid complex) helps pull the epiglottis open after swallowing.

Group 2: Cord Position (Abduction vs. Adduction)

This group controls the width of the Rima Glottidis (the air gap).

Muscle	Action	Mechanism
Posterior Cricoarytenoid (PCA)	Abducts (Opens)	Rotates arytenoids outward
Lateral Cricoarytenoid (LCA)	Adducts (Closes)	Rotates arytenoids inward
Transverse Arytenoid	Adducts (Closes)	Pulls arytenoids together

Warning: The Posterior Cricoarytenoid is the **only** muscle that abducts (opens) the vocal cords. It is the only skeletal muscle in the body that acts as an abductor of the airway.

Clinical Focus: Because PCA is the sole abductor, its paralysis causes airway obstruction. Bilateral PCA paralysis is a life-threatening emergency because the adductors remain active, clamping the air shut.

Group 3: Cord Tension (Pitch and Tone)

These muscles alter the length and tension of the vocal ligaments to modulate pitch.

Muscle	Action	Mechanism
Cricothyroid (CT)	Tenses	Tilt thyroid forward/down
Vocalis	Tenses (Anterior) / Relaxes (Posterior)	Fine modulation
Thyroarytenoid (TA)	Relaxes	Shortens vocal cord

Exam Focus:

- **Cricothyroid:** The "Tuning Fork" of the larynx. It tenses the cords for high pitch.
- **Vocalis:** The "Violin Finger." It allows fine-tuning of pitch during speech.
- **Thyroarytenoid:** The "Relaxer." It shortens the cord for low pitch.

Detailed Mechanism: The Cricothyroid Muscle

The Cricothyroid is unique among intrinsic muscles.

1. **Location:** It is the **only** intrinsic muscle lying on the **external** surface of the larynx.
2. **Innervation:** External Laryngeal Nerve (branch of Vagus CN X).
3. **Action:**
 - Origin: Anterolateral arch of Cricoid.
 - Insertion: Inferior cornu and lower border of Thyroid lamina.
 - Contraction tilts the thyroid cartilage downward and forward relative to the cricoid.
 - **Result:** The vocal cords are stretched, lengthened, and tensed.

Exam Answer: The Cricothyroid muscle increases pitch by stretching the vocal cords. It is supplied by the External Laryngeal Nerve.

Detailed Mechanism: The Vocalis Muscle

The Vocalis is the medial part of the Thyroarytenoid.

1. **Location:** Lies within the vocal fold, lateral to the vocal ligament.
2. **Innervation:** Recurrent Laryngeal Nerve.
3. **Action:**
 - Tenses the anterior part of the vocal ligament.
 - Relaxes the posterior part.
 - Allows for fine modulation of voice pitch (segmental tension).

Key Point: Vocalis paralysis affects voice quality and pitch control but does not cause airway obstruction.

Detailed Mechanism: The Posterior Cricoarytenoid (PCA)

The PCA is the most clinically significant intrinsic muscle.

1. **Location:** Posterior surface of the cricoid lamina.
2. **Innervation:** Recurrent Laryngeal Nerve.
3. **Action:**
 - Pulls the muscular process of the arytenoid backward and laterally.
 - This rotation moves the vocal process laterally.
 - **Result:** Abduction of vocal cords (widens the glottis).

Clinical Focus:

- **Unilateral PCA Paralysis:** The affected cord rests in a paramedian position. The patient may have a weak voice but breathes normally.
- **Bilateral PCA Paralysis:** Both cords are fixed in adduction. The airway is blocked. **Emergency tracheostomy** is required.

Innervation Summary Table

Muscle Group	Nerve Supply	Clinical Consequence of Lesion
Cricothyroid	External Laryngeal Nerve	Loss of high pitch, weak voice
All Other Intrinsic Muscles	Recurrent Laryngeal Nerve	Hoarseness (unilateral), Airway obstruction (bilateral)

Must Remember:

- **External Laryngeal Nerve** -> Cricothyroid only.
- **Recurrent Laryngeal Nerve** -> All other intrinsic muscles.

Exam Trap: Students often confuse the nerves. Remember: "Cricothyroid is External; Rest are Recurrent."

Functional Summary Checklist

Use this checklist to verify understanding:

- ☐ **Extrinsic Elevators:** Thyrohyoid, Palatopharyngeus, Salpingopharyngeus, Stylopharyngeus.
- ☐ **Extrinsic Depressor:** Sternothyroid.
- ☐ **Inlet Closers:** Oblique Arytenoid, Aryepiglotticus.
- ☐ **Inlet Opener:** Thyroepiglotticus.
- ☐ **Abductor (Airway):** Posterior Cricoarytenoid (PCA).
- ☐ **Adductors (Closure):** Lateral Cricoarytenoid, Transverse Arytenoid.
- ☐ **Tensors (Pitch):** Cricothyroid, Vocalis.
- ☐ **Relaxer (Low Pitch):** Thyroarytenoid.
- ☐ **Unpaired Muscle:** Transverse Arytenoid.
- ☐ **External Muscle:** Cricothyroid.

Answer Line: The intrinsic muscles of the larynx are functionally grouped into inlet controllers (Oblique/Aryepiglotticus), cord positioners (PCA abducts, LCA/Transverse adduct), and tensioners (CT/Vocalis tense, TA relaxes).

Intrinsic Muscles: The Safety Muscles (Abductors)

Definition/Core Idea The posterior cricoarytenoid (PCA) muscles are the sole abductors of the vocal cords in the larynx. They are uniquely critical because they are the only intrinsic laryngeal muscles responsible for opening the airway. All other intrinsic muscles tend to adduct (close) the glottis or tense the cords.

Clinical Focus: Because the PCA is the only muscle that opens the airway, its paralysis is a life-threatening emergency. Bilateral paralysis causes immediate airway obstruction.

Anatomy and Mechanics of Abduction

Origin and Insertion

- **Origin:** Posterior surface of the cricoid lamina, lateral to the median ridge.
- **Insertion:** Back of the muscular process of the arytenoid cartilage.
- **Shape:** Triangular muscle.

Mechanism of Action The PCA acts through a specific rotational mechanism at the cricoarytenoid joint:

1. **Contraction:** The muscle pulls the muscular process of the arytenoid cartilage medially.
2. **Rotation:** This pull rotates the arytenoid cartilage around a vertical axis.
3. **Abduction:** As the muscular process moves medially, the vocal process (which is on the opposite side of the arytenoid base) swings laterally.
4. **Result:** The vocal cords separate, widening the rima glottidis into a diamond-shaped opening.

Key Point: This is the only muscle in the body that abducts the vocal cords. If the PCA fails, the adductor muscles (lateral cricoarytenoid and transverse arytenoid) dominate, pulling the cords into a closed position.

Innervation

- **Nerve Supply:** Recurrent laryngeal nerve (branch of the vagus nerve, CN X).
- **Clinical Note:** The recurrent laryngeal nerve is vulnerable during thyroid surgery. Injury to this nerve directly compromises the PCA function.

Clinical Significance: The "Safety Muscle" Concept

Why is it called the Safety Muscle? The PCA is termed the "safety muscle" because it is the primary guardian of the airway. Its sole job is to ensure air can enter the lungs.

Unilateral Paralysis

- **Cause:** Damage to one recurrent laryngeal nerve (e.g., during thyroidectomy).

- **Effect:** The affected vocal cord rests in a paramedian position (between abduction and adduction).
- **Symptoms:** Hoarseness is present, but airway obstruction is usually minimal because the healthy contralateral cord can still abduct sufficiently for breathing.

Bilateral Paralysis

- **Cause:** Damage to both recurrent laryngeal nerves.
- **Effect:** Both vocal cords lie in a paramedian position. The airway is significantly narrowed.
- **Symptoms:** Severe difficulty in breathing (dyspnea) and loss of phonation. This is an airway emergency.

Exam Answer: The posterior cricoarytenoid is the only abductor of the vocal cords. It is supplied by the recurrent laryngeal nerve. Bilateral paralysis of this muscle causes airway obstruction because the adductor muscles pull the cords into a closed position, blocking airflow.

Summary Checklist for Exams

- **Muscle:** Posterior Cricothyroid (PCA).
- **Action:** Abduction of vocal cords (opening the glottis).
- **Origin:** Posterior cricoid lamina.
- **Insertion:** Muscular process of arytenoid.
- **Nerve:** Recurrent laryngeal nerve.
- **Unique Feature:** Only abductor in the larynx; all other intrinsic muscles adduct or tense.
- **Clinical Trap:** Bilateral PCA paralysis = Airway emergency (stridor/respiratory distress).

Must Remember: Do not confuse PCA with the cricothyroid. The cricothyroid tenses the cords (pitch control) and is supplied by the external laryngeal nerve. The PCA opens the airway and is supplied by the recurrent laryngeal nerve.

Intrinsic Muscles: The Adductors (Lateral Cricothyroid, Transverse Arytenoid)

These two muscles are the primary active closers of the glottis. While the posterior cricoarytenoid opens the airway, the lateral cricoarytenoid and transverse arytenoid work together to bring the vocal cords together for phonation and airway protection.

Functional Classification and Grouping

The intrinsic muscles of the larynx are divided by action. Understanding this grouping is essential for exam answers regarding glottic dynamics.

- **Abductors (Openers):** Posterior cricoarytenoid (the only abductor).
- **Adductors (Closers):** Lateral cricoarytenoid and Transverse arytenoid.
- **Tensors/Relaxers:** Cricothyroid, Vocalis, Thyroarytenoid.

Exam Focus: All intrinsic muscles adduct the vocal cords except the posterior cricoarytenoid, which abducts them.

Lateral Cricothyroid

This is a paired triangular muscle responsible for the rotational component of adduction.

- **Origin:** Lateral part of the upper border of the cricoid arch.
- **Insertion:** Front of the muscular process of the arytenoid cartilage.
- **Action:** Adducts the vocal cords.

Mechanism of Action:

When the lateral cricothyroid contracts, it pulls the muscular process of the arytenoid cartilage forward. Because the arytenoid rotates around a vertical axis at the cricothyroid joint, this forward pull on the posterior aspect causes the vocal process (and the attached vocal cord) to swing medially into the midline.

Key Point: The lateral cricothyroid works in conjunction with the transverse arytenoid to achieve complete glottic closure.

Transverse Arytenoid

This is the only unpaired intrinsic muscle of the larynx. It provides the sliding component of adduction.

- **Type:** Unpaired, rectangular muscle.
- **Origin/Insertion:** Connects the posterior surfaces of the two arytenoid cartilages. It runs transversely from the posterior surface of one arytenoid to the posterior surface of the other.
- **Action:** Adducts the vocal cords.

Mechanism of Action:

Contraction of the transverse arytenoid pulls the two arytenoid cartilages toward each other. This gliding movement narrows the posterior (intercartilaginous) part of the rima glottidis.

Exam Focus: Transverse arytenoid is the only unpaired intrinsic muscle of the larynx.

Comparative Summary: The Adductors

To write a precise exam answer, distinguish the specific mechanical contribution of each adductor.

Feature	Lateral Cricoarytenoid	Transverse Arytenoid
Paired Status	Paired	Unpaired
Origin	Lateral upper border of cricoid arch	Posterior surface of one arytenoid
Insertion	Front of muscular process of arytenoid	Posterior surface of opposite arytenoid
Primary Motion	Rotatory (swings vocal process medially)	Gliding (pulls arytenoids together)
Glottic Effect	Closes intermembranous part	Closes intercartilaginous part

Must Remember: The lateral cricoarytenoid rotates the arytenoid; the transverse arytenoid slides the arytenoids together. Both actions are necessary for complete glottic closure.

Clinical and Exam Relevance

Glottic Closure and Airway Protection The coordinated action of these adductors is vital for:

1. **Phonation:** Bringing the vocal cords together to vibrate against exhaled air.
2. **Swallowing:** Closing the glottis to prevent aspiration of food or liquid into the trachea.

Paralysis Patterns

- **Unilateral Recurrent Laryngeal Nerve Damage:** The affected vocal cord rests in a paramedian position. The adductors on the healthy side can still pull the paralyzed cord toward the midline to some extent, allowing for near-normal phonation, though breathiness may occur.
- **Bilateral Recurrent Laryngeal Nerve Damage:** Both vocal cords are paralyzed in the paramedian position. This causes severe difficulty in breathing (stridor) because the posterior cricoarytenoids (abductors) cannot open the airway. The adductors remain relatively tight, keeping the glottis narrowed. This is a medical emergency requiring airway management.

Answer Line: The lateral cricoarytenoid and transverse arytenoid are the adductor muscles of the larynx; the former rotates the arytenoid cartilage to swing the vocal process medially, while the latter pulls the arytenoids together to close the posterior glottis.

Intrinsic Muscles: The Tensors (Cricothyroid, Vocalis, Thyroarytenoid)

These three muscles control the tension of the vocal cords, which is the primary mechanism for determining pitch and voice modulation. While other intrinsic muscles open or close the glottis, these tensors prepare the cords to vibrate.

Must Remember: The Cricothyroid is the only intrinsic laryngeal muscle supplied by the External Laryngeal Nerve. All others are supplied by the Recurrent Laryngeal Nerve.

Cricothyroid: The Pitch Setter

The cricothyroid is unique because it is the only intrinsic muscle located on the external surface of the larynx. It acts as the primary "tuning fork" of the larynx.

- **Origin:** Anterolateral aspect of the arch of the cricoid cartilage.
- **Insertion:** Fibres pass backwards and upwards to insert into the inferior cornu and the adjacent lower border of the lamina of the thyroid cartilage.
- **Shape:** Small, fan-shaped muscle.
- **Nerve Supply:** External laryngeal nerve.

Mechanism of Action When the cricothyroid contracts, it tilts the thyroid cartilage downwards and forwards relative to the cricoid cartilage. This movement is often described as the "visor mechanism" (like lifting the visor of a knight's helmet).

- **Effect:** This tilting lengthens and tenses the vocal cords.
- **Result:** Increased tension raises the pitch of the voice.
- **Secondary Action:** It also assists in the adduction of the vocal cords.

Exam Focus: The cricothyroid is critical for high-pitched phonation. Paralysis of this muscle (due to external laryngeal nerve injury) causes a significant alteration in voice tone, particularly noticeable in singers who lose their ability to produce high notes.

Vocalis: The Modulator

The vocalis is the medial, detached part of the thyroarytenoid muscle. It lies within the vocal fold, just lateral and cranial to the vocal ligament.

- **Origin:** Posterior surface of the angle of the thyroid cartilage and the anterior part of the vocal ligament.
- **Insertion:** Lateral surface of the vocal process of the arytenoid cartilage.
- **Nerve Supply:** Recurrent laryngeal nerve.

Function The vocalis does not act on the entire vocal cord uniformly. It provides segmental tension.

- **Anterior Fibres:** Contraction tenses the anterior part of the vocal ligament.
- **Posterior Fibres:** Contraction relaxes the posterior part of the vocal ligament.
- **Analogy:** This action is compared to the fingers of a violinist pressing on strings. By altering tension at specific points, the vocalis allows for fine modulation of voice quality and pitch, rather than just global tension changes.

Thyroarytenoid: The Relaxed State

The thyroarytenoid is the broad muscle that forms the bulk of the vocal cord. It is the antagonist to the cricothyroid in terms of tension.

- **Origin:** Posterior aspect of the angle of the thyroid cartilage.
- **Insertion:** Anterolateral surface of the arytenoid cartilage.
- **Nerve Supply:** Recurrent laryngeal nerve.

Function

- **Primary Action:** Relaxes the vocal cords.
- **Mechanism:** By contracting, it shortens the distance between the thyroid angle and the arytenoid, reducing tension on the vocal ligament.
- **Clinical Context:** This muscle helps return the larynx to a relaxed state after phonation.

Comparative Summary of Tensor Muscles

Feature	Cricothyroid	Vocalis	Thyroarytenoid
Primary Action	Tenses cords (High Pitch)	Modulates tension (Fine Pitch)	Relaxes cords (Low Pitch)
Location	External surface of larynx	Within the vocal fold	Bulk of the vocal cord
Nerve Supply	External Laryngeal Nerve	Recurrent Laryngeal Nerve	Recurrent Laryngeal Nerve
Key Analogy	Tuning fork / Visor	Violinist's fingers	Relaxer
Origin	Cricoid cartilage	Thyroid angle / Vocal ligament	Thyroid angle
Insertion	Thyroid cartilage (inferior cornu)	Vocal process of arytenoid	Arytenoid cartilage

Key Point: The cricothyroid is the sole muscle supplied by the external laryngeal nerve. This is a high-yield exam distinction because injury to this nerve (e.g., during thyroid surgery) specifically impairs high-pitch phonation, whereas recurrent laryngeal nerve injury affects all other intrinsic muscles, leading to hoarseness or airway obstruction.

Exam Answer: When asked about pitch control, state that pitch is determined by vocal cord tension. The **Cricothyroid** increases tension (high pitch) via the external laryngeal nerve, while the **Thyroarytenoid** decreases tension (low pitch) via the recurrent laryngeal nerve. The **Vocalis** provides fine-tuning modulation.

Intrinsic Muscles: Inlet Openers/Closers

The laryngeal inlet is the primary gateway between the pharynx and the laryngeal cavity. Its protection during swallowing is critical to prevent aspiration. While the posterior cricoarytenoids manage the glottis for breathing, the muscles controlling the inlet manage the seal between the upper aerodigestive tract and the lower airway.

Functional Classification

The intrinsic muscles controlling the inlet are functionally divided into two opposing groups: those that close the inlet (predominant during swallowing) and those that open it (predominant during respiration and speech).

- **Closers (Constrictors):** Oblique arytenoid, Aryepiglotticus.
- **Openers (Dilators):** Thyroepiglotticus.

Key Point: The inlet is normally kept open by the elastic recoil of the epiglottis and the tone of the thyroepiglotticus. It is closed actively only during the pharyngeal phase of swallowing.

The Closers: Oblique Arytenoid and Aryepiglotticus

These two muscles act together to narrow the laryngeal inlet. They are anatomically continuous and functionally synergistic.

- **Oblique Arytenoid:**
 - **Origin:** Muscular process of one arytenoid cartilage.
 - **Insertion:** Apex of the opposite arytenoid cartilage.
 - **Action:** Pulls the arytenoid cartilages together, narrowing the inlet.
- **Aryepiglotticus:**
 - **Origin:** Muscular process of the arytenoid cartilage.
 - **Insertion:** Margin of the epiglottis.
 - **Relation:** It is essentially the upper, more lateral fibers of the oblique arytenoid that extend into the aryepiglottic fold.
 - **Action:** Pulls the epiglottis downward and medially, helping to seal the inlet against the arytenoids.

Exam Focus: The aryepiglotticus is not a separate muscle with a distinct origin/insertion from the oblique arytenoid in many anatomical descriptions; rather, it is the continuation of the oblique arytenoid fibers into the aryepiglottic fold.

The Opener: Thyroepiglotticus

This muscle opposes the action of the closers. It is often considered a functional part of the thyroarytenoid muscle.

- **Origin:** Posterior aspect of the angle of the thyroid cartilage.
- **Insertion:** Margin of the epiglottis.
- **Relation:** It represents the upper fibers of the thyroarytenoid muscle that curve upward into the aryepiglottic fold.
- **Action:** Pulls the epiglottis upward and laterally, widening the laryngeal inlet.

Must Remember: The thyroepiglotticus opens the inlet. This is counter-intuitive because the thyroarytenoid (its parent muscle) generally adducts the vocal cords. However, its upper fibers specifically target the epiglottis to dilate the inlet.

Anatomical Context: The Aryepiglottic Fold

The aryepiglottic fold is a critical landmark. It contains the muscular fibers of the oblique arytenoid, aryepiglotticus, and thyroepiglotticus.

- **Boundaries:** Extends from the lateral margin of the epiglottis to the apex of the arytenoid cartilage.
- **Contents:**
 - Muscular fibers (Oblique arytenoid, Aryepiglotticus, Thyroepiglotticus).
 - Corniculate cartilages (at the apex of the arytenoid).
 - Cuneiform cartilages (embedded in the fold, above the corniculate).

Clinical Focus: During direct laryngoscopy, the aryepiglottic folds are prominent landmarks. Pathology in this region (e.g., tumors) can impair the closing mechanism, leading to aspiration.

Functional Summary Table

Muscle	Group	Action on Inlet	Primary Context
Oblique Arytenoid	Closer	Closes/Narrows	Swallowing
Aryepiglotticus	Closer	Closes/Narrows	Swallowing
Thyroepiglotticus	Opener	Opens/Widens	Respiration, Speech

Exam Answer Frame

Question: Describe the muscles controlling the laryngeal inlet.

Answer Line: The laryngeal inlet is controlled by the oblique arytenoid and aryepiglotticus (closers) and the thyroepiglotticus (opener).

- Key Points:**
1. **Closers:** The oblique arytenoid crosses from one arytenoid to the other. The aryepiglotticus is its continuation into the aryepiglottic fold, attaching to the epiglottis. Together, they pull the epiglottis down and the arytenoids together to seal the inlet during swallowing.
 2. **Opener:** The thyroepiglotticus is the upper part of the thyroarytenoid muscle. It attaches to the epiglottis and pulls it upward to open the inlet during breathing.
 3. **Clinical Relevance:** Failure of the closers leads to aspiration; failure of the opener is less common but would restrict airway dilation.

Warning: Do not confuse the thyroepiglotticus with the thyrohyoid muscle. The thyrohyoid is an extrinsic muscle that elevates the larynx; the thyroepiglotticus is intrinsic and modulates the inlet size.

Laryngeal Cavity: Inlet and Subdivisions

Definition and Clinical Significance of the Inlet

The laryngeal cavity is the internal hollow space of the larynx, continuous above with the laryngopharynx and below with the laryngotrachea. Its upper opening is called the **laryngeal inlet** (or *aditus laryngis*). This inlet serves as the gateway between the respiratory/digestive tracts above and the lower airway below.

Clinical Focus: The laryngeal inlet is a critical site for airway management. Because it is a narrow passage, obstruction here (e.g., by edema, foreign body, or tumor) can be rapidly fatal. Understanding its boundaries is essential for identifying sites of pathology and for procedural landmarks during intubation.

Boundaries of the Laryngeal Inlet

The inlet is bounded by specific anatomical structures that form a protective and functional frame. You must identify these three components:

- **Anteriorly:** The posterior surface of the epiglottis.
- **Laterally:** The aryepiglottic folds on both sides. These folds extend from the lateral margin of the epiglottis to the apex of the arytenoid cartilages.
- **Posteriorly:** The interarytenoid fold, which connects the posterior margins of the two arytenoid cartilages.

Key Point: The aryepiglottic folds contain the aryepiglotticus muscles (part of the oblique arytenoids), which can contract to narrow the inlet during swallowing, helping to protect the airway.

Subdivisions of the Laryngeal Cavity

The laryngeal cavity is divided into three distinct compartments by two pairs of folds. This division is crucial for understanding laryngeal function and pathology.

1. **Vestibule (Supraglottic Cavity):** The uppermost compartment, located above the vestibular folds.
2. **Ventricle (Intermediate Cavity):** The middle compartment, located between the vestibular and vocal folds.
3. **Infraglottic Cavity:** The lower compartment, located below the vocal folds, leading into the trachea.

The Vestibular (False) Folds

The upper pair of folds are the **vestibular folds**, commonly known as the **false vocal cords**.

- **Structure:** They are formed by the vestibular ligament (derived from the lower edge of the quadrangular membrane) covered by mucous membrane.
- **Function:** They do not participate in phonation. Instead, they protect the true vocal cords and help close the laryngeal inlet during swallowing.
- **Appearance:** In laryngoscopy, they appear red and are located lateral to the true cords.

Exam Focus: Do not confuse vestibular folds with vocal folds. Vestibular folds are "false" because they do not produce sound. They are thicker and contain more glandular tissue.

The Vocal (True) Folds

The lower pair of folds are the **vocal folds**, commonly known as the **true vocal cords**.

- **Structure:** They are formed by the vocal ligament (derived from the upper edge of the cricovocal membrane/conus elasticus) and the vocalis muscle.
- **Function:** They are the primary structures for phonation. Their vibration produces sound when air passes through.
- **Appearance:** In laryngoscopy, they appear pearly white and are located medial to the false cords.

Warning: The vocal folds have a different epithelial lining (stratified squamous) compared to the rest of the larynx (ciliated columnar) to withstand friction during vibration. This difference is clinically significant for cancer risk and histology questions.

The Ventricle of the Larynx

The space between the vestibular and vocal folds is the **ventricle of the larynx** (also called the sinus of the larynx).

- **Structure:** It is a narrow, slit-like cavity.
- **Contents:** It contains the **saccule of the larynx**, a small diverticulum of the mucous membrane that extends upward between the folds. The saccule contains mucous glands that lubricate the vocal folds.
- **Clinical Relevance:** Inflammation or cysts in the saccule can lead to a laryngocele, a condition where air-filled sacs form in this region.

Summary Table: Laryngeal Compartments

Compartment	Location	Boundaries	Key Structures
Vestibule (Supraglottic)	Upper	Above vestibular folds	Aryepiglottic folds, false cords
Ventricle (Intermediate)	Middle	Between vestibular and vocal folds	Saccule of larynx, mucous glands
Infraglottic	Lower	Below vocal folds	True vocal cords, leading to trachea

Exam Answer: The laryngeal cavity is divided into three parts: the supraglottic vestibule (above false cords), the intermediate ventricle (between false and true cords), and the infraglottic cavity (below true cords). The division is created by the vestibular (false) and vocal (true) folds.

The Folds: Vestibular (False) vs Vocal (True)

Definition and Core Concept

The laryngeal cavity contains two pairs of mucosal folds projecting from the lateral walls toward the midline. These are the **vestibular folds** (false vocal cords) superiorly and the **vocal folds** (true vocal cords) inferiorly. They are separated by a lateral recess called the **sinus of the larynx** (ventricle).

- **Vestibular Folds (False):** Located in the supraglottic region. They do not participate in phonation. Their primary role is protective, helping to close the laryngeal inlet during swallowing.
- **Vocal Folds (True):** Located in the glottic region. They are the primary organs of phonation. They vibrate to produce sound when air passes through the rima glottidis.

Key Point: The distinction between "true" and "false" is functional. Only the true folds vibrate to create speech.

Structural Composition

The two pairs of folds differ significantly in their underlying support and tissue composition.

1. Vestibular Fold (False Cord)

- **Support:** The **vestibular ligament**. This is a thickened lower edge of the **quadrangular membrane**.
- **Extent:** Extends from the posterior surface of the thyroid cartilage (above the cricothyroid membrane) to the lateral surface of the arytenoid cartilage.
- **Tissue:** Covered by mucosa containing submucosa. This loose connective tissue allows the fold to swell.

1. Vocal Fold (True Cord)

- **Support:** The **vocal ligament**. This is the free upper edge of the **cricovocal membrane** (conus elasticus).
- **Extent:** Extends from the posterior surface of the thyroid cartilage (at the angle) to the vocal process of the arytenoid cartilage.
- **Tissue:** Lacks a submucosal layer. The mucosa is tightly adherent to the vocal ligament. This tight adherence is essential for efficient vibration.

Histological Differences

The histology of the coverings explains the functional and clinical differences between the folds.

- **Vestibular Folds:** Covered by **ciliated columnar epithelium** (respiratory epithelium). They contain a rich submucosa with mucous glands. This allows them to trap particles and swell if inflamed.
- **Vocal Folds:** Covered by **non-keratinized stratified squamous epithelium**. This tissue is resistant to friction from vibration. Crucially, there is **no submucosa** and **no mucous glands** on the true vocal cords. The mucosa is directly attached to the vocal ligament.

Exam Focus: The absence of submucosa on the true vocal cords is a high-yield anatomical fact. It explains why edema does not occur on the true cords, unlike the false cords.

The Saccule of the Larynx

The **sinus of the larynx** (ventricle) is the horizontal space between the vestibular and vocal folds.

- **Location:** It lies between the false and true vocal cords.
- **Saccule:** The posterior part of the sinus extends upward as a small blind pouch called the **saccule of the larynx**.
- **Contents:** The saccule contains mucous glands.
- **Clinical Relevance:** Inflammation or infection can lead to a **laryngocele** (dilation of the saccule), which may fill with air or mucus and cause a neck swelling or voice changes.

Comparative Summary Table

Use this table for rapid revision and exam comparison.

Feature	Vestibular Folds (False)	Vocal Folds (True)
Alternative Name	False vocal cords	True vocal cords
Location	Supraglottic (Upper)	Glottic (Lower)
Underlying Ligament	Vestibular ligament	Vocal ligament
Membrane Origin	Quadrangular membrane	Cricovocal membrane (Conus Elasticus)
Primary Function	Protection (swallowing)	Phonation (speech)
Epithelium	Ciliated Columnar	Stratified Squamous (Non-keratinized)
Submucosa	Present (loose tissue)	Absent (tight adherence)
Glands	Present	Absent
Edema Potential	High (swells easily)	Low (does not swell)
Appearance	Red, fleshy	Pearly white

Clinical and Exam Relevance

Clinical Focus: Edema and Airway Obstruction Because the vestibular folds have a loose submucosa, they are prone to edema (swelling) during allergic reactions or infections. This can rapidly compromise the airway. The true vocal cords do not swell because the mucosa is tightly bound to the ligament.

Clinical Focus: Laryngoscopy Appearance During laryngoscopy:

- The **vestibular folds** appear red and fleshy.
- The **vocal folds** appear pearly white and lie medial to the vestibular folds.
- The **sinus of the larynx** is visible as a deep recess between them.

Must Remember:

- **Vestibular = Protection** (False cords close the inlet).
- **Vocal = Phonation** (True cords vibrate for sound).
- **Saccule** is the extension of the sinus between the folds.
- **True cords** have no submucosa; **False cords** do.

Exam Answer Frame:

Question: Differentiate between vestibular and vocal folds. Answer:

1. **Location:** Vestibular are superior (false); Vocal are inferior (true).
2. **Function:** Vestibular protect the larynx during swallowing; Vocal produce sound.
3. **Histology:** Vestibular have ciliated columnar epithelium and submucosa; Vocal have stratified squamous epithelium and no submucosa.
4. **Support:** Vestibular rest on the vestibular ligament; Vocal rest on the vocal ligament.
5. **Clinical:** Vestibular folds can swell (edema); Vocal folds cannot.

Rima Glottidis: Boundaries and Shapes

Definition: The rima glottidis is the narrowest anteroposterior cleft within the laryngeal cavity. It represents the critical functional aperture for airflow and phonation.

Key Point: The rima glottidis is the narrowest part of the entire laryngeal cavity. Any obstruction here causes immediate respiratory distress.

Boundaries of the Rima Glottidis

The boundaries are defined by the cartilaginous framework and the mucosal folds. You must identify these structures during laryngoscopy or dissection.

- **Anteriorly:** The angle of the thyroid cartilage.
- **Posteriorly:** The interarytenoid folds of the mucous membrane.
- **Laterally (on each side):**
 - **Anterior 3/5:** The vocal fold (true vocal cord).
 - **Posterior 2/5:** The vocal process of the arytenoid cartilage.

Exam Focus: Do not confuse the vestibular fold (false cord) with the vocal fold. The rima glottidis is bounded *only* by the true vocal folds and vocal processes. The space above the true folds is the ventricle; the space above that is the vestibule.

Subdivisions of the Rima Glottidis

The rima glottidis is anatomically divided into two distinct regions based on the underlying support structures.

1. Intermembranous Part (Anterior 3/5):

- Located between the two vocal ligaments (vocal cords).
- This is the primary vibrating segment during phonation.
- It is bounded by the elastic vocal ligaments.

1. Intercartilaginous Part (Posterior 2/5):

- Located between the vocal processes of the two arytenoid cartilages.
- This region does not vibrate significantly but opens and closes to regulate airflow.
- It is bounded by the mucosal interarytenoid folds.

Must Remember: The division is not arbitrary. The anterior 3/5 is membrane-supported (intermembranous), while the posterior 2/5 is cartilage-supported (intercartilaginous).

Shapes of the Rima Glottidis

The shape of the rima glottidis is dynamic. It changes based on the action of the intrinsic laryngeal muscles, specifically the cricoarytenoid joints. You must know these four shapes for exam questions on laryngoscopy views.

State / Action	Shape of Rima Glottidis	Anatomical Mechanism
Quiet Breathing	Pentagonal	The intermembranous part is triangular, and the intercartilaginous part is rectangular. The overall outline is pentagonal.
Full Inspiration	Diamond	The posterior cricoarytenoid muscles abduct the vocal cords. The vocal processes move laterally, widening the glottis into a diamond shape to maximize airflow.
High-Pitched Voice	Linear Chink	Both the intermembranous and intercartilaginous parts adduct tightly. The vocal cords come together to form a narrow linear slit, increasing tension for high-frequency vibration.
Whispering	Inverted Funnel	The intermembranous part is highly adducted (closed), but the intercartilaginous part remains open (separated). This creates a triangular gap posteriorly, allowing turbulent air to pass without full cord vibration.

Exam Answer: When asked to describe the shape of the glottis during inspiration, state: "Diamond-shaped due to abduction of the vocal cords by the posterior cricoarytenoid muscles."

Clinical Focus: In bilateral recurrent laryngeal nerve paralysis, the vocal cords lie in the paramedian position. This narrows the rima glottidis, causing stridor and respiratory difficulty because the abductors (posterior cricoarytenoids) are paralyzed, and the adductors pull the cords toward the midline.

Dimensions and Sex Differences

The size of the rima glottidis correlates with sex and age due to the size of the larynx.

- **Adult Males:** Anteroposterior diameter is approximately **24 mm**.
- **Adult Females:** Anteroposterior diameter is approximately **16 mm**.

Key Point: The larger diameter in males contributes to the lower pitch of the male voice, as the longer and thicker vocal folds vibrate at a lower frequency.

Answer Line: The rima glottidis is the narrowest part of the larynx, bounded anteriorly by the thyroid angle, posteriorly by the interarytenoid folds, and laterally by the vocal folds (anterior 3/5) and vocal processes of arytenoids (posterior 2/5).

Mechanism of Phonation

Phonation is the production of voice through the vibration of the true vocal cords. The larynx functions as a wind instrument, where the lungs provide the power source, the larynx acts as the vibrating reed, and the pharynx, mouth, and nose serve as the resonating chambers.

Key Point: The primary function of the larynx is protection; phonation is a secondary, evolutionarily later function.

The Wind Instrument Analogy

To understand phonation, visualize the larynx as a musical wind instrument (like a clarinet or trumpet).

- **Power Source:** The lungs provide a steady stream of air (pneumotoric energy).
- **Vibrating Element:** The true vocal cords act as the reed or valve.
- **Resonating Chambers:** The pharynx, oral cavity, and nasal cavity modify the raw sound into recognizable speech.

Without the coordinated action of these three components, no voice is produced. The raw sound generated at the larynx is known as the **laryngeal tone**.

Steps of Sound Production

Sound production is a cyclic process involving three distinct phases. This cycle repeats hundreds of times per second during speech.

1. Glottal Closure (Adduction):

The vocal cords are brought together in the midline by the adductor muscles (lateral cricoarytenoid and transverse arytenoid). This closes the rima glottidis, stopping the airflow from the lungs.

1. Subglottic Pressure Build-up:

With the glottis closed, air continues to be pushed from the lungs. This builds up pressure below the vocal cords (subglottic pressure).

1. Glottal Opening (Abduction) and Vibration:

When the subglottic pressure exceeds the resistance of the closed cords, the vocal cords are blown apart. Air rushes through the gap, producing a sound pulse. The Bernoulli effect and elastic recoil of the cords then pull them back together, restarting the cycle.

Exam Answer: Phonation occurs when adducted vocal cords vibrate due to the cyclic build-up and release of subglottic air pressure.

Determinants of Pitch and Loudness

The quality of the voice is determined by two main physical properties: frequency (pitch) and amplitude (loudness).

1. Pitch (Frequency)

Pitch refers to the highness or lowness of the voice. It is determined by the **frequency** of vibration of the vocal cords.

- **High Pitch:** Fast vibrations (high frequency). Occurs when cords are short, thick, and relaxed.
- **Low Pitch:** Slow vibrations (low frequency). Occurs when cords are long, thin, and tense.

Mechanism of Pitch Control:

- **To raise pitch:** The cricothyroid muscle contracts. This pulls the thyroid cartilage forward and downward, stretching and tensing the vocal cords. Tense cords vibrate faster.
- **To lower pitch:** The thyroarytenoid (vocalis) muscle relaxes or contracts slightly to shorten and thicken the cords.

Exam Focus: The **Cricothyroid muscle** is the primary tensor of the vocal cords. It is often called the "tuning fork of the larynx" because it adjusts the tension for pitch.

2. Loudness (Amplitude)

Loudness refers to the volume of the voice. It is determined by the **amplitude** of the vibration and the force of the exhaled air.

- **High Loudness:** Large amplitude of vibration + strong expiratory force.
- **Low Loudness:** Small amplitude of vibration + gentle expiratory force.

Clinical Focus: Male voices are typically louder but lower pitched than female voices. This is because male vocal cords are longer and heavier. They vibrate with greater amplitude (louder) but at a lower frequency (lower pitch).

Summary Table: Pitch vs. Loudness

Feature	Determinant	Physical Basis	Primary Muscle Action
Pitch	Frequency	Speed of vibration	Cricothyroid (tenses cords for high pitch)
Loudness	Amplitude	Force of vibration	Respiratory muscles (force of air)

Must Remember:

- **Pitch** depends on the **tension** of the vocal cords (controlled by cricothyroid muscle).
- **Loudness** depends on the **amplitude** of vibration (controlled by expiratory force).

Clinical Correlation: Vocal Nodules

Vocal nodules are benign, bilateral swellings on the vocal cords, often seen in teachers, singers, or individuals who shout.

- **Location:** They occur at the junction of the anterior one-third and posterior two-thirds of the vocal cords.
- **Cause:** This is the area of maximum contact and friction during vibration.
- **Result:** Chronic irritation leads to inflammatory nodules, causing hoarseness and voice fatigue.

Warning: Unilateral hoarseness or voice change lasting more than 2 weeks requires investigation to rule out laryngeal cancer, especially in smokers.

Nerve Supply: Motor and Sensory Maps

The larynx possesses a precise dual innervation system that separates motor control from sensory perception based on anatomical boundaries. Mastering these maps is critical for understanding clinical presentations of nerve injury, such as hoarseness or airway obstruction.

Core Nerves and Their Origins

The laryngeal innervation derives from three distinct branches of the vagus nerve (CN X). You must distinguish between the **Superior Laryngeal Nerve** and the **Recurrent Laryngeal Nerve** branches.

- **Superior Laryngeal Nerve:** Divides into two terminal branches:
- **Internal Laryngeal Nerve:** Purely sensory.
- **External Laryngeal Nerve:** Purely motor.
- **Recurrent Laryngeal Nerve:** Mixed nerve (motor and sensory) that loops around the subclavian artery (right) or aortic arch (left) before ascending to the larynx.

Key Point: The internal laryngeal nerve pierces the thyrohyoid membrane to enter the larynx. This is a key anatomical landmark for localization.

Sensory Distribution: The Vocal Cord Boundary

Sensory supply is divided by the vocal cords. This boundary is a high-yield exam concept because it dictates the clinical presentation of pathology.

1. Above the Vocal Cords (Supraglottic):

- **Nerve:** Internal Laryngeal Nerve.
- **Area:** Epiglottis, arytenoids, false vocal cords, and the upper part of the laryngeal vestibule.
- **Clinical Significance:** Stimulation here triggers the cough reflex and the gag reflex. A lesion here may cause silent aspiration because the patient cannot feel food entering the airway.

1. Below the Vocal Cords (Infraglottic):

- **Nerve:** Recurrent Laryngeal Nerve.
- **Area:** True vocal cords (lower surface), subglottic region, and trachea.
- **Clinical Significance:** Sensation here is less prominent in daily function but relevant in trauma or intubation injury.

Exam Focus: The vocal cords themselves are largely insensate on their free edges, but the mucosa immediately above and below is densely innervated. The "boundary" is the vocal fold.

Motor Distribution: The Cricothyroid Exception

Motor supply follows a simple rule: **All intrinsic muscles are supplied by the Recurrent Laryngeal Nerve, except the Cricothyroid.**

- **Recurrent Laryngeal Nerve:**
 - Supplies all intrinsic muscles: Posterior cricoarytenoid, Lateral cricoarytenoid, Transverse arytenoid, Thyroarytenoid, Vocalis, and Oblique arytenoid.
 - **Function:** Controls abduction, adduction, and fine tension modulation of the vocal cords.
- **External Laryngeal Nerve:**
 - Supplies the **Cricothyroid muscle** only.
 - **Function:** Tenses the vocal cords (increases pitch).
 - **Why this matters:** The cricothyroid is the only muscle on the *external* surface of the larynx. It is the primary tensor of the vocal folds.

Must Remember: The Cricothyroid is the "tuning fork" of the larynx. It sets the fundamental tension. All other muscles refine the pitch and shape.

Clinical Correlation: Nerve Injury Patterns

Understanding nerve injury requires knowing which muscles are paralyzed and how the vocal cords rest.

Nerve Injured	Motor Deficit	Vocal Cord Position	Clinical Presentation
Unilateral RLN	Paralysis of all intrinsic muscles on one side (except CT)	Paramedian (slightly adducted)	Hoarse voice. The healthy cord can often compensate by crossing midline.
Bilateral RLN	Paralysis of all intrinsic muscles on both sides	Paramedian (fixed adducted position)	Airway Obstruction. The cords cannot abduct to breathe. This is a surgical emergency.
External LN	Paralysis of Cricothyroid only	Normal resting position	Loss of high-pitch voice. Voice becomes flat and monotonous. Common in thyroid surgery.
Internal LN	Sensory loss above cords	No motor deficit	Loss of cough reflex. Risk of silent aspiration.

Warning: In bilateral RLN injury, the vocal cords are often stuck in a paramedian position that *narrows* the airway, leading to stridor and respiratory distress. This is distinct from the "cadaveric position" where cords are fully abducted.

The Cadaveric Position

When all laryngeal nerves are severed (or in death), the vocal cords assume a specific resting state known as the cadaveric position.

- **Muscles involved:** All intrinsic muscles are paralyzed.
- **Result:** The vocal cords lie in a **paramedian position** but with the cricothyroid relaxed.
- **Airway:** The rima glottidis is narrow.
- **Exam Trap:** Do not confuse this with the position in bilateral RLN injury. In bilateral RLN injury, the cords are also paramedian, but the mechanism is unopposed adduction by the cricothyroid (if intact) or balanced paralysis. In total denervation, the lack of tension from the cricothyroid allows the cords to relax into a slightly more open, but still narrow, position.

Exam Answer: The cadaveric position of the vocal cords is a paramedian position with the rima glottidis reduced to a narrow slit, resulting from paralysis of all intrinsic laryngeal muscles.

Summary Checklist for Revision

- [] **Internal Laryngeal Nerve:** Sensory above vocal cords.
- [] **External Laryngeal Nerve:** Motor to Cricothyroid (tenses cords).
- [] **Recurrent Laryngeal Nerve:** Motor to all other intrinsic muscles + Sensory below vocal cords.
- [] **Unilateral RLN injury:** Hoarseness, paramedian cord.
- [] **Bilateral RLN injury:** Airway obstruction, paramedian cords.

- [] **External LN injury:** Loss of high pitch, monotonous voice.

Clinical Focus: During thyroidectomy, the external laryngeal nerve is at risk near the superior thyroid artery. The recurrent laryngeal nerve is at risk near the inferior thyroid artery. Protecting both requires careful dissection in these vascular zones.

Vascular Supply: Arteries, Veins, Lymphatics

Arterial Supply

The blood supply to the larynx is derived from two main sources: the superior laryngeal artery and the inferior laryngeal artery. These vessels provide a rich anastomotic network within the laryngeal mucosa.

Superior Laryngeal Artery

- **Origin:** Arises from the thyroid artery.
- **Entry Point:** Pierces the thyrohyoid membrane along with the internal laryngeal nerve.
- **Distribution:** Supplies the mucous membrane and muscles of the upper part of the larynx (above the vocal cords).

Inferior Laryngeal Artery

- **Origin:** Arises from the inferior thyroid artery.
- **Entry Point:** Enters the larynx through the cricothyroid membrane.
- **Distribution:** Supplies the lower part of the larynx (below the vocal cords).

Key Point: The superior and inferior laryngeal arteries anastomose freely within the larynx. This collateral circulation is clinically significant during surgical procedures, as it helps maintain tissue viability even if one supply is compromised.

Venous Drainage

Venous drainage follows the arterial supply pattern but drains into different venous systems.

- **Superior Laryngeal Veins:** Drain the upper part of the larynx. They accompany the superior laryngeal artery and drain into the internal jugular vein.
- **Inferior Laryngeal Veins:** Drain the lower part of the larynx. They accompany the inferior laryngeal artery and drain into the thyroid ima vein or the brachiocephalic vein.

Exam Focus: Note that while the arterial supply is paired (superior/inferior), the venous drainage is often grouped simply as "superior" and "inferior" laryngeal veins without the specific artery name attached in some texts, but the anatomical correlation remains strict.

Lymphatic Drainage

Lymphatic drainage of the larynx is critical for understanding the spread of laryngeal cancer. The drainage pattern is sharply divided by the level of the vocal cords.

Above the Vocal Cords (Supraglottic Region)

- **Primary Drainage:** Lymphatics drain directly to the **upper deep cervical lymph nodes**.
- **Secondary Drainage:** From there, they may spread to lower deep cervical nodes.
- **Clinical Implication:** Because this region has rich lymphatic supply, cancers here (e.g., in the epiglottis or false cords) tend to spread early to cervical nodes. Prognosis is generally poorer due to early metastasis.

Below the Vocal Cords (Infraglottic Region)

- **Primary Drainage:** Lymphatics drain to the **prelaryngeal (anterior) and pretracheal nodes**.
- **Secondary Drainage:** These nodes then drain into the **lower deep cervical lymph nodes**.
- **Clinical Implication:** This region has sparse lymphatic supply. Cancers here (e.g., in the true vocal cords) are less likely to spread to lymph nodes early. Prognosis is generally better if detected early.

Must Remember: The vocal cords themselves have very few lymphatics. This is why glottic cancers (cancer of the true vocal cords) rarely metastasize to lymph nodes in the early stages.

Clinical Focus: Lymphatic Drainage and Cancer Prognosis

The difference in lymphatic drainage above and below the vocal cords is a high-yield exam topic and a crucial clinical distinction.

- **Glottic Cancer (True Cords):** Rarely metastasizes early. Good prognosis.
- **Supraglottic Cancer (Above Cords):** Metastasizes early to upper deep cervical nodes. Poorer prognosis.
- **Infraglottic Cancer (Below Cords):** Metastasizes to prelaryngeal/pretracheal nodes, then lower deep cervical nodes.

Exam Answer: When asked about the prognosis of laryngeal cancer, always mention the lymphatic density. Glottic cancers have a better prognosis because the vocal cords are relatively avascular and have sparse lymphatics, delaying metastasis.

Summary Table: Vascular Supply of the Larynx

Feature	Superior Laryngeal	Inferior Laryngeal
Artery Origin	Thyroid artery	Inferior thyroid artery
Entry Point	Thyrohyoid membrane	Cricothyroid membrane
Area Supplied	Above vocal cords	Below vocal cords
Venous Drainage	Internal jugular vein	Thyroid ima / Brachiocephalic vein
Lymphatic Drainage	Upper deep cervical nodes	Prelaryngeal/Pretracheal -> Lower deep cervical nodes

Warning: Do not confuse the entry points. The superior laryngeal vessels enter *above* the thyroid cartilage (via thyrohyoid membrane), while the inferior laryngeal vessels enter *below* the thyroid cartilage (via cricothyroid membrane). This is a common exam trap.

Source Note: The source text explicitly states that lymphatics above the vocal cords drain to upper deep cervical nodes, while those below drain to prelaryngeal/pretracheal nodes and then to lower deep cervical nodes. This distinction is vital for clinical staging.

Clinical Correlations: Nerve Injuries

Core Concept: The Airway vs. The Voice

Injury to the laryngeal nerves alters the position and tension of the vocal cords. The clinical presentation depends on which nerve is damaged and whether the injury is unilateral or bilateral.

The key to understanding these injuries is knowing which muscles control abduction (opening) and which control adduction (closing).

- **Abduction (Opening):** Posterior cricoarytenoid muscles only. These are the "safety muscles" because they keep the airway open.
- **Adduction (Closing):** Lateral cricoarytenoid, Transverse arytenoid, Thyroarytenoid, Vocalis.
- **Tension:** Cricothyroid (External Laryngeal Nerve) and Vocalis/Thyroarytenoid (Recurrent Laryngeal Nerve).

When a nerve is cut, the muscles it supplies relax. The unopposed action of the remaining intact muscles pulls the vocal cord into a specific resting position.

Unilateral Recurrent Laryngeal Nerve (RLN) Injury

The recurrent laryngeal nerve supplies all intrinsic laryngeal muscles except the cricothyroid.

Mechanism of Injury This is commonly caused by accidental damage during thyroidectomy or other neck surgeries.

Cord Position The affected vocal cord assumes a **paramedian position**.

- The posterior cricoarytenoid (abductor) is paralyzed.
- The adductor muscles (lateral cricoarytenoid, transverse arytenoid) are also paralyzed, but the **cricothyroid muscle** remains intact (supplied by the external laryngeal nerve).
- The intact cricothyroid muscle tenses the cord and pulls it slightly toward the midline.
- Result: The cord lies between the midline and the lateral wall, not fully abducted.

Clinical Symptoms

- **Hoarseness:** The paralyzed cord does not vibrate. The healthy cord vibrates against the stationary paralyzed cord, producing a weak, breathy voice.
- **Compensation:** In unilateral injury, the healthy cord can often cross the midline to meet the paralyzed cord. Therefore, phonation is not completely lost, but it is significantly weakened.

- **Aspiration Risk:** Minimal, as the paramedian position still provides some protection during swallowing.

Exam Answer Frame

Question: What is the position of the vocal cord in unilateral RLN paralysis?

Answer: The vocal cord lies in a **paramedian position**.

Reason: The posterior cricoarytenoid (abductor) is paralyzed. The cricothyroid (supplied by external laryngeal nerve) remains intact and pulls the cord slightly toward the midline. The voice is hoarse but present.

Bilateral Recurrent Laryngeal Nerve (RLN) Injury

This is a medical emergency due to airway compromise.

Mechanism of Injury Usually results from bilateral thyroidectomy or severe neck trauma.

Cord Position Both vocal cords lie in the **paramedian position**.

Clinical Symptoms

- **Airway Obstruction:** This is the primary danger. The posterior cricoarytenoids (the only abductors) are paralyzed on both sides. The cords cannot open to allow air entry.
- **Stridor:** A high-pitched breathing sound caused by air forcing its way through the narrowed glottic opening.
- **Dyspnea:** Severe difficulty in breathing.
- **Voice:** Phonation is severely impaired or absent because both cords are paralyzed and cannot vibrate effectively.

Clinical Focus: Bilateral RLN injury is an airway emergency. Immediate tracheostomy may be required to secure the airway.

Must Remember: The posterior cricoarytenoid muscles are the **only** muscles that abduct the vocal cords. Their paralysis causes airway obstruction.

External Laryngeal Nerve (ELN) Injury

The external laryngeal nerve supplies the cricothyroid muscle.

Mechanism of Injury Can occur during thyroidectomy if the nerve is damaged near the superior thyroid artery.

Cord Position The vocal cord remains in its normal resting position (slightly abducted) because the adductors and abductors are balanced. However, the cord is **relaxed** because the cricothyroid (tenser) is paralyzed.

Clinical Symptoms

- **Voice Fatigue:** The patient cannot sustain high-pitched sounds.
- **Loss of Pitch:** The voice becomes weak and monotonous.
- **No Hoarseness:** Unlike RLN injury, the cord moves normally, so there is no breathy voice.
- **Significance in Singers:** This injury is devastating for professional singers who rely on fine pitch control. The cricothyroid is often called the "tuning fork of the larynx."

Exam Focus: Differentiate RLN vs. ELN injury.

- **RLN Injury:** Hoarseness, paramedian cord.
- **ELN Injury:** Voice fatigue, loss of pitch, normal cord position.

Cadaveric Position of the Vocal Cords

This term describes the position of the vocal cords when **both** the recurrent laryngeal nerve and the external laryngeal nerve are damaged on both sides.

Mechanism

- All intrinsic laryngeal muscles are paralyzed.
- The posterior cricoarytenoid (abductor) is paralyzed.
- The cricothyroid (tenser/puller) is paralyzed.
- The adductor muscles are also paralyzed.

Cord Position The vocal cords lie in an **intermediate position** between abduction and adduction. They are slightly abducted, fixed in place.

Clinical Consequence

- **Airway:** The airway is patent (open) because the cords are not adducted. This is actually safer for breathing than bilateral RLN injury.
- **Voice:** Complete loss of phonation.
- **Aspiration:** High risk of aspiration during swallowing because the cords cannot close to protect the airway.

Key Point: The "cadaveric position" is an intermediate, fixed position resulting from paralysis of all laryngeal muscles. It preserves the airway but destroys the voice and protective reflexes.

Summary Table: Nerve Injuries

Nerve Injured	Muscle Affected	Cord Position	Primary Symptom
Unilateral RLN	Posterior cricoarytenoid (and others)	Paramedian	Hoarseness
Bilateral RLN	Posterior cricoarytenoids (both)	Paramedian (both)	Airway obstruction (Stridor)
External LN	Cricothyroid	Normal (relaxed)	Voice fatigue, loss of pitch
Bilateral RLN + ELN	All intrinsic muscles	Intermediate (Cadaveric)	Loss of voice, aspiration risk

Exam Answer: Differentiating RLN and ELN Injury

Question: How do you clinically differentiate between recurrent laryngeal nerve and external laryngeal nerve injury?

Answer Line: Unilateral RLN injury causes hoarseness with a paramedian cord position, while external laryngeal nerve injury causes voice fatigue and loss of pitch with a normal cord position.

Explanation:

- 1. RLN Injury:** Affects the posterior cricoarytenoid (abductor) and adductor muscles. The cord assumes a paramedian position. The voice is hoarse and breathy.
- 2. ELN Injury:** Affects only the cricothyroid muscle (tenser). The cord moves normally but cannot tense. The voice is weak, fatigues quickly, and lacks high pitch. There is no hoarseness.

Warning: Do not confuse the paramedian position of RLN injury with the intermediate position of cadaveric paralysis. In RLN injury, the cricothyroid is still working, pulling the cord medially. In cadaveric paralysis, the cricothyroid is also paralyzed, leaving the cord in a neutral, slightly abducted position.

Clinical Correlations: Obstruction, Laryngoceles, and Nodules

Laryngeal Obstruction and the Heimlich Maneuver

Laryngeal obstruction occurs when a foreign body blocks the laryngeal inlet or glottis, preventing air entry. This is a medical emergency that can lead to rapid asphyxia if the airway is not cleared.

Mechanism of the Heimlich Maneuver The Heimlich maneuver is the primary first-aid technique for dislodging a foreign body lodged in the larynx.

1. **Action:** The rescuer applies a sudden, forceful upward pressure on the patient's abdomen (just below the rib cage).
2. **Anatomical Effect:** This pressure pushes the diaphragm upward sharply.
3. **Result:** The upward movement of the diaphragm compresses the lungs, forcing a rapid column of air out of the trachea.
4. **Dislodgement:** This high-pressure jet of air acts like an artificial cough, blowing the obstructing object out of the larynx.

Clinical Focus: The maneuver works by creating an artificial respiration force to expel the foreign body. It is critical to recognize that if the obstruction is complete, no air can enter, and the patient cannot speak or breathe.

Risk Factor: Alcohol Intoxication Alcohol consumption significantly increases the risk of choking on food.

- **Reason:** Alcohol depresses the central nervous system.
- **Mechanism:** It blunts the protective laryngeal reflexes, specifically the cough reflex and the closure of the glottis during swallowing.
- **Outcome:** A person who is intoxicated may swallow food while the laryngeal inlet is not properly protected, leading to aspiration or obstruction.

Exam Focus: When explaining choking risk, always link alcohol to the depression of protective reflexes (cough/glottic closure), not just "slurred speech" or coordination.

Laryngocele

A laryngocele is a pathological dilation or herniation of the saccule of the larynx.

Anatomical Basis

- **Origin:** The saccule is a small mucous-filled pouch located in the ventricle of the larynx, between the vestibular (false) and vocal (true) folds.
- **Pathology:** Increased pressure within the larynx or weakness in the wall of the saccule causes it to expand.
- **Herniation:** The dilated saccule pushes outward through the thyrohyoid membrane.

Clinical Presentation

- **Location:** The swelling appears in the lateral aspect of the neck, near the thyroid cartilage.
- **Character:** It is often soft and compressible.
- **Contents:** It may contain air (if the mucous membrane is intact and communicates with the laryngeal cavity) or mucus (if obstructed).

Key Point: A laryngocele is essentially a herniation of the saccule of the larynx through the thyrohyoid membrane. It is a rare but distinct clinical entity that requires surgical correction if it causes airway compromise.

Vocal Nodules (Singer's Nodules)

Vocal nodules are benign, non-neoplastic growths that form on the true vocal cords. They are commonly referred to as "singer's nodules" due to their association with voice overuse.

Location

- **Specific Site:** They occur at the junction of the anterior one-third and the posterior two-thirds of the true vocal cords.
- **Why this site?** This is the point of maximum impact and vibration during phonation. The repeated trauma from vocal cord collision causes reactive tissue hyperplasia.

Etiology

- **Cause:** Chronic vocal abuse or misuse (e.g., shouting, singing with poor technique, excessive talking).
- **Pathology:** The friction creates small callus-like thickenings on the mucosa of the vocal folds.

Clinical Relevance

- **Symptoms:** Hoarseness, breathiness, or loss of voice range.
- **Management:** Voice rest and speech therapy are the primary treatments. Surgical removal is rarely needed and only considered if conservative measures fail.

Must Remember: The location of vocal nodules is a high-yield exam fact. Always specify the **junction of the anterior 1/3 and posterior 2/3** of the vocal cords. This is the point of greatest vibratory stress.

Comparison of Clinical Pathologies

Feature	Laryngeal Obstruction	Laryngocele	Vocal Nodules
Primary Cause	Foreign body aspiration	Herniation of saccule	Vocal abuse/overuse
Anatomical Site	Laryngeal inlet/ Glottis	Saccule (ventricle)	True vocal cords (Ant 1/3- Post 2/3 junction)
Key Mechanism	Physical blockage of airway	Pressure-induced dilation	Reactive hyperplasia from friction
Immediate Risk	Asphyxia (Life-threatening)	Airway compression (if large)	Hoarseness/Dysphonia
First Aid/ Management	Heimlich Maneuver	Surgical excision	Voice rest/Speech therapy

Exam Answer: When asked about the cause of hoarseness in a singer, think of vocal nodules located at the junction of the anterior and posterior thirds of the vocal cords. When asked about the mechanism of the Heimlich maneuver, describe the upward pressure on the diaphragm creating an air jet to dislodge the object.

High-Yield Tables and Golden Facts

This section consolidates the most frequently tested facts, clinical correlations, and structural comparisons of the larynx. Use this for rapid revision and to verify your understanding of key distinctions.

Must Remember: These facts are high-yield for exams. Memorize the specific names, positions, and clinical implications listed below.

Golden Facts Checklist

- **Foundation stone of larynx:** Cricoid cartilage.
- **Tuning fork of larynx:** Cricothyroid muscle (tenses vocal cords).
- **Safety muscle of larynx:** Posterior cricoarytenoid (only abductor; prevents asphyxia).
- **Narrowest part of laryngeal cavity:** Rima glottidis.
- **Oil can of larynx:** Saccule of larynx (lubricates vocal cords).
- **Cadaveric position of vocal cords:** Paramedian position (due to bilateral recurrent laryngeal nerve damage).
- **Commonest site for laryngeal cancer:** Glottis (True vocal cords).
- **Most common cause of laryngeal obstruction:** Foreign body (food) causing choking.
- **Heimlich maneuver indication:** Laryngeal obstruction by foreign body.

- **Cartilage that ossifies after 25 years:** Thyroid, Cricoid, and basal part of Arytenoid (Hyaline cartilage).
- **Cartilage that does NOT ossify:** Epiglottis, Corniculate, Cuneiform, and apices of Arytenoid (Elastic cartilage).
- **Nerve supply to Cricothyroid muscle:** External laryngeal nerve.
- **Nerve supply to all other intrinsic muscles:** Recurrent laryngeal nerve.
- **Sensory supply above vocal cords:** Internal laryngeal nerve.
- **Sensory supply below vocal cords:** Recurrent laryngeal nerve.
- **Lymphatic drainage above vocal cords:** Upper deep cervical nodes (low risk of spread).
- **Lymphatic drainage below vocal cords:** Pretracheal/Paratracheal nodes (high risk of spread due to rich lymphatics).

Comparison Tables

Table 1: Classification of Laryngeal Cartilages

Feature	Thyroid	Cricoid	Epiglottis	Arytenoid	Corniculate	Cuneiform
Number	Unpaired	Unpaired	Unpaired	Paired	Paired	Paired
Type	Hyaline	Hyaline	Elastic	Hyaline (base) / Elastic (apex)	Elastic	Elastic
Ossification	Yes (>25 yrs)	Yes (>25 yrs)	No	Base: Yes; Apex: No	No	No
Key Landmark	Adam's apple (males)	Foundation stone	Leaf-shaped	Pyramidal	At apex of arytenoid	Above corniculate
Clinical Note	Prominent in males	Ring shape protects airway	Protects inlet during swallowing	Articulates with cricoid	Non-functional joint	Non-functional joint

Table 2: Intrinsic Muscles of the Larynx

Muscle	Action on Vocal Cords	Nerve Supply	Clinical/Exam Significance
Posterior Cricoarytenoid	Abducts (Opens)	Recurrent Laryngeal	Safety muscle. Paralysis causes airway obstruction.
Lateral Cricoarytenoid	Adducts (Closes)	Recurrent Laryngeal	Helps in phonation.
Transverse Arytenoid	Adducts (Closes)	Recurrent Laryngeal	Unpaired muscle.
Cricothyroid	Tenses	External Laryngeal	Tuning fork. Paralysis causes weak voice.
Vocalis	Tenses/ Modulates	Recurrent Laryngeal	Fine-tunes pitch (like violin fingers).
Thyroarytenoid	Relaxes	Recurrent Laryngeal	Shortens vocal cords.
Oblique Arytenoid	Closes inlet	Recurrent Laryngeal	Swallowing reflex.

Table 3: Nerve Supply and Clinical Correlations

Nerve	Type	Distribution	Clinical Deficit
Internal Laryngeal	Sensory	Supraglottic mucosa (above cords)	Loss of cough reflex; aspiration risk.
External Laryngeal	Motor	Cricothyroid muscle	Weak voice; loss of high pitch.
Recurrent Laryngeal	Motor/ Sensory	All intrinsic muscles (except CT) + Subglottic mucosa	Hoarseness (unilateral); Asphyxia (bilateral).

Exam Focus: Be precise with "Above" vs "Below" the vocal cords.

- **Arteries:** Superior laryngeal (above), Inferior laryngeal (below).
- **Lymphatics:** Above cords drain to upper deep cervical nodes (sparse lymphatics, good prognosis). Below cords drain to pretracheal nodes (rich lymphatics, poor prognosis).

Clinical Scenarios and Exam Hooks

- 1. Nerve Injury Patterns

- **Unilateral Recurrent Laryngeal Nerve (RLN) Damage:**

- Vocal cord lies in **paramedian position**.
- Patient presents with **hoarseness**.
- The healthy cord can often compensate by crossing the midline.

- **Bilateral RLN Damage:**

- Both cords lie in **paramedian position**.
- **No phonation** (silent voice).
- **Respiratory difficulty** due to narrow glottis.

- **External Laryngeal Nerve Damage:**

- Loss of tension in vocal cords.
- Voice becomes weak and monotonous; singers notice this first.

- **Combined RLN + External Laryngeal Nerve Damage:**

- Vocal cords in **cadaveric position** (midline, abducted).
- Complete paralysis of intrinsic muscles.

- **2. Laryngeal Obstruction**

- **Cause:** Foreign body (food) lodging in the laryngeal inlet.
- **Reflex:** Explosive coughing due to sensitivity of supraglottic mucosa (Internal laryngeal nerve).
- **Emergency: Heimlich maneuver.**
- Mechanism: Abdominal thrusts increase intrathoracic pressure.
- Result: Air blast dislodges the foreign body.
- **Warning:** If reflex is absent (e.g., alcohol intoxication, neurological lesion), obstruction can lead to rapid asphyxia.

- **3. Laryngocele**

- **Definition:** Herniation of the saccule of the larynx.
- **Cause:** Increased intralaryngeal pressure (e.g., trumpet players, glass blowers).
- **Types:**
- **Internal:** Within the larynx.
- **External:** Herniates through the thyrohyoid membrane.

- **4. Vocal Nodules**

- **Location:** Junction of anterior 1/3 and posterior 2/3 of vocal cords.
- **Cause:** Overuse of voice (teachers, singers).
- **Appearance:** Bilateral, symmetrical, soft/reddish initially, becoming greyish/whitish.
- **Exam Answer:** "Singer's nodules" or "Screamer's nodules".

- **5. Laryngoscopy Views**

- **Direct Laryngoscopy:** View of laryngeal inlet, epiglottis, aryepiglottic folds, and vocal cords.
- **Key Landmarks:**
- **Rima Glottidis:** Opening between true vocal cords.
- **Rima Vestibuli:** Opening between false vocal cords.
- **Ventricles:** Spaces between true and false cords.

Must Remember: Do not confuse the **vestibular folds** (false cords) with the **vocal folds** (true cords).

- **Vestibular folds:** Protect inlet during swallowing; prevent air escape when holding breath.
- **Vocal folds:** Produce sound; narrowest part of cavity.

Quick Revision Summary

- **Location:** Anterior neck, C3-C6.
- **Functions:** Phonation, Respiration, Protection, Deglutition.
- **Cartilages:** 9 total (4 unpaired, 5 paired).
- **Joints:** Cricothyroid, Cricoarytenoid, Arytenocorniculate.
- **Muscles:** 10 intrinsic muscles (7 pairs, 1 unpaired).
- **Nerves:** Internal (Sensory), External (Motor to CT), Recurrent (Motor to rest + Sensory below).
- **Blood Supply:** Superior/Inferior Laryngeal arteries.
- **Lymphatics:** Above cords -> Upper deep cervical; Below cords -> Pretracheal.

Exam Answer: When asked for the "most important muscle for respiration," answer **Posterior Cricoarytenoid**. When asked for the "most important muscle for phonation," answer **Cricothyroid** (tension) and **Vocalis** (modulation).

High-Yield Tables and Lists

Cartilage Classification and Characteristics

Key Point: The larynx contains 9 cartilages. Distinguishing between hyaline and elastic types is critical for understanding ossification patterns and radiological visibility.

Cartilage Name	Type	Number	Key Anatomical Features & Clinical Notes
Thyroid	Hyaline	1 (Unpaired)	Most prominent; forms "Adam's apple" (laryngeal prominence). Angle is 90° in males (acute) and 120° in females (obtuse). Acts as a shield.
Cricoid	Hyaline	1 (Unpaired)	Signet-shaped ring; narrow anterior arch, broad posterior lamina. Located at C6 level. Known as the "foundation stone" of the larynx.
Epiglottis	Elastic	1 (Unpaired)	Leaf-shaped; upper end free, lower end attached to thyroid angle via thyroepiglottic ligament. Protects airway during swallowing.
Arytenoid	Hyaline (base) / Elastic (apex)	2 (Paired)	Pyramidal shape. Articulates with cricoid. Has muscular process (for muscle attachment) and vocal process (for vocal ligament attachment).
Corniculate	Elastic	2 (Paired)	Small conical nodules; articulate with apices of arytenoids. Located in posterior parts of aryepiglottic folds.
Cuneiform	Elastic	2 (Paired)	Tiny rod-shaped cartilages; lie above corniculate cartilages in aryepiglottic folds.

Exam Focus: Hyaline cartilages (Thyroid, Cricoid, basal Arytenoids) tend to ossify after age 25 and are visible on radiographs. Elastic cartilages (Epiglottis, Corniculate, Cuneiform, apical Arytenoids) do not ossify.

Laryngeal Joints and Movements

Key Point: Understanding joint types and movements explains how vocal cord abduction and adduction occur mechanically.

Joint Name	Type	Articulating Structures	Movements Permitted	Clinical/Functional Significance
Cricothyroid	Plane Synovial	Inferior cornu of Thyroid cartilage & side of Cricoid cartilage	1. Rotatory (around transverse axis) 2. Gliding (limited)	Rotation tenses the vocal cords. The recurrent laryngeal nerve enters the larynx very close to this joint.
Cricoarytenoid	Plane Synovial	Base of Arytenoid & lateral upper border of Cricoid lamina	1. Rotatory (around vertical axis) 2. Gliding (towards/away)	Rotation abducts/adducts vocal cords. Gliding closes/opens the posterior glottis.
Arytenocorniculate	Synovial	Arytenoid & Corniculate cartilages	None	Of no functional significance.

Warning: Injury to the cricothyroid joint region is dangerous due to the proximity of the recurrent laryngeal nerve and superior laryngeal vessels.

Intrinsic Muscles of the Larynx

Must Remember: All intrinsic muscles are paired **except** the Transverse Arytenoid. All intrinsic muscles are adductors **except** the Posterior Cricoarytenoids (which are abductors).

Muscle Name	Action	Nerve Supply	Functional Role
Posterior Cricoarytenoid	Abduction of vocal cords	Recurrent Laryngeal Nerve	Safety Muscle: Opens the glottis for breathing. Only abductor in the larynx.
Lateral Cricoarytenoid	Adduction of vocal cords	Recurrent Laryngeal Nerve	Closes the glottis; aids in phonation.
Transverse Arytenoid	Adduction of vocal cords	Recurrent Laryngeal Nerve	Closes the posterior glottis. Unpaired muscle.
Oblique Arytenoid	Closes laryngeal inlet	Recurrent Laryngeal Nerve	Constricts the laryngeal inlet; assists in swallowing.
Aryepiglotticus	Closes laryngeal inlet	Recurrent Laryngeal Nerve	Part of the aryepiglottic fold; protects airway during swallowing.
Thyroepiglotticus	Closes laryngeal inlet	Recurrent Laryngeal Nerve	Fibers of oblique arytenoid that attach to epiglottis.
Cricothyroid	Tenses vocal cords	External Laryngeal Nerve	Raises pitch of voice. "Tuning fork" of the larynx.
Thyroarytenoid	Relaxes vocal cords	Recurrent Laryngeal Nerve	Shortens and relaxes cords; lowers pitch.
Vocalis	Modulates/ Tenses cords	Recurrent Laryngeal Nerve	Fine-tunes pitch and loudness. "Modulators of voice."

Exam Focus: The Cricothyroid is the **only** intrinsic muscle supplied by the **External Laryngeal Nerve**. All others are supplied by the Recurrent Laryngeal Nerve.

Rima Glottidis Shapes During Function

Key Point: The shape of the glottis changes dynamically based on respiratory and phonatory needs. This is a high-yield visual concept for exams.

Physiological State	Shape of Rima Glottidis	Mechanism
Quiet Breathing	Pentagonal	Intermembranous part is triangular; intercartilaginous part is rectangular.
Full Inspiration	Diamond-shaped	Vocal cords abduct widely to maximize airflow.
High-Pitched Voice	Linear Chink	Both intermembranous and intercartilaginous parts adduct tightly.
Whispering	Inverted Funnel	Intermembranous part is highly adducted; intercartilaginous part remains separated by a triangular gap.

Clinical Focus: In unilateral vocal cord paralysis, the glottis appears deviated toward the paralyzed side during abduction attempts.

Laryngeal Innervation and Clinical Lesions

Warning: Nerve injuries are a common exam topic. Distinguish between sensory and motor deficits.

Nerve	Type	Distribution	Clinical Deficit if Injured
Internal Laryngeal	Sensory	Mucosa above vocal cords (supraglottic)	Loss of cough reflex; aspiration risk.
External Laryngeal	Motor	Cricothyroid muscle	Loss of ability to produce high-pitched voice; voice becomes hoarse and weak.
Recurrent Laryngeal	Motor & Sensory	All intrinsic muscles (except CT) + Mucosa below cords	Unilateral: Hoarseness, breathy voice. Bilateral: Airway obstruction (stridor), emergency tracheostomy required.

Exam Answer: The **Posterior Cricoarytenoid** is the only muscle that abducts the vocal cords. Bilateral injury to the recurrent laryngeal nerves causes bilateral abduction failure, leading to airway obstruction.

Lymphatic Drainage of the Larynx

Key Point: The vocal cords have a "lymphatic watershed" area. Drainage patterns differ significantly above and below this line.

Region	Drainage Pathway	Final Destination
Above Vocal Cords (Supraglottic)	Drain to Superior Deep Cervical Lymph Nodes (pre-epiglottic nodes first)	Jugular chain nodes
Below Vocal Cords (Infraglottic)	Pierce cricothyroid membrane → Prelaryngeal & Pretracheal Nodes	Lower Deep Cervical Lymph Nodes (posterior-inferior group)
Vocal Cords (Glottic)	Sparse lymphatics; drain to Prelaryngeal Nodes	Lower Deep Cervical Lymph Nodes

Clinical Focus: Laryngeal cancer spreads differently based on location. Glottic cancers (on the cords) spread late due to sparse lymphatics, whereas supraglottic cancers spread early to cervical nodes.

Phonation Mechanism Checklist

Must Remember: Phonation is a multi-step process involving aerodynamics, acoustics, and articulation.

- Adduction:** Vocal cords are brought together (adducted) by adductor muscles.
- Subglottic Pressure:** Exhaled air from lungs increases pressure below the cords (via abdominal/intercostal muscles).
- Opening:** Air pressure forces the cords apart, releasing air in small puffs.
- Vibration:** Elastic recoil and Bernoulli effect cause the cords to vibrate rapidly.
- Sound Production:** Vibration produces a fundamental tone.
- Amplification:** Resonating chambers (mouth, pharynx, nose, esophagus) amplify the sound.
- Articulation:** Lips, tongue, palate, and teeth modify the sound into speech.

Key Point:

- **Loudness** depends on the **amplitude** of vibration.
- **Pitch** depends on the **frequency** of vibration.
- Males have longer cords → lower frequency → lower pitch.
- Females have shorter cords → higher frequency → higher pitch.

Applied / Clinical / Exam Relevance

The larynx is a high-yield topic for anatomy exams because it integrates complex skeletal mechanics, precise neurovascular supply, and critical clinical emergencies. Examiners focus heavily on the **mechanism of nerve injuries**, the **distinction between true and false vocal cords**, and the **emergency management of airway obstruction**.

Must Remember: The primary function of the larynx is **protection** of the lower respiratory tract, not phonation. Phonation is a secondary, evolved function.

1. Airway Obstruction and the Heimlich Maneuver

Clinical Focus: Foreign body aspiration is a life-threatening emergency often seen in adults who consume alcohol or have impaired swallowing reflexes.

- **Site of Lodgement:** The narrowest part of the adult larynx is the **Rima Glottidis**. This is where foreign bodies (e.g., meat, bones) typically lodge.
- **Mechanism of Choking:**
 1. Alcohol impairs the cough reflex and reduces the ability to chew properly.
 2. A large bolus enters the laryngeal inlet.
 3. The epiglottis fails to seal the glottis effectively.
 4. The object lodges at the Rima Glottidis, causing complete asphyxia.
- **Heimlich Maneuver (First Aid):**
 - **Action:** Abdominal thrusts (pressure on the epigastrium).
 - **Mechanism:** Compresses the abdomen → increases intrathoracic pressure → forces air out of the lungs → creates an "artificial cough" → dislodges the foreign body.
 - **Exam Trap:** Do not confuse this with CPR. This is specifically for **upper airway obstruction** in a conscious or semi-conscious patient.

Exam Answer: The narrowest part of the larynx is the **Rima Glottidis**. Foreign bodies lodge here because it is the physiological bottleneck of the laryngeal cavity.

2. Nerve Injuries: The "Paralysis" Spectrum

Laryngeal nerve injuries are classic exam questions. You must distinguish between the **Recurrent Laryngeal Nerve (RLN)** and the **External Laryngeal Nerve**.

A. Recurrent Laryngeal Nerve (RLN) Injury

The RLN supplies all intrinsic muscles **except** the Cricothyroid.

Type of Injury	Clinical Presentation	Mechanism
Unilateral RLN	Hoarseness	One vocal cord is paralyzed in the paramedian position (midway between adducted and abducted). The healthy cord cannot meet the paralyzed cord, causing incomplete glottic closure.
Bilateral RLN	Stridor / Asphyxia	Both cords are paralyzed in the paramedian position . The glottis is narrowed to a slit. Air cannot enter, causing severe respiratory distress.

Warning: Bilateral RLN injury is a medical emergency requiring immediate tracheostomy. The airway is compromised, not the voice.

B. External Laryngeal Nerve Injury

This nerve supplies the **Cricothyroid muscle** (the tensor of the vocal cords).

- **Presentation:** **Loss of high-pitched voice** and fatigue during prolonged speaking.
- **Mechanism:** The Cricothyroid is responsible for tensing the vocal cords to produce high notes. Without it, the patient cannot raise the pitch of their voice.
- **Exam Trap:** Patients may complain of "weak voice" but do not have hoarseness like RLN injury. The voice is monotonous and low.

Must Remember:

- **RLN Injury** = Hoarseness (Unilateral) / Asphyxia (Bilateral).
- **External Laryngeal Injury** = Loss of high pitch.

3. Laryngeal Cancer and Lymphatic Spread

Clinical Focus: Squamous Cell Carcinoma (SCC) accounts for 90–95% of laryngeal cancers. The site of the tumor determines the prognosis and lymphatic spread.

- **Commonest Site: Glottic region** (True vocal cords).
- **Why Glottic Cancer has Good Prognosis:**
 - The true vocal cords have **very few lymphatics**.
 - Early metastasis is rare.
 - Symptoms (hoarseness) appear early, leading to early diagnosis.
- **Why Supraglottic Cancer has Poor Prognosis:**
 - The supraglottic region (above vocal cords) has **rich lymphatic drainage**.
 - Cancer spreads early to deep cervical nodes.
 - Symptoms (pain, dysphagia) appear late.

Exam Answer: The most common site of laryngeal cancer is the **Glottic region** (True vocal cords). It has a better prognosis because the vocal cords are **lymph-poor**, delaying metastasis.

4. Vocal Nodules ("Singer's Nodules")

Clinical Focus: Common in teachers, singers, and speakers who overuse their voice.

- **Pathology:** Inflammatory nodules (polyps) on the vocal cords.
- **Site:** Junction of the **anterior one-third** and **posterior two-thirds** of the vocal cords.
- **Reason:** This is the point of **maximum contact and friction** during vibration.
- **Presentation:** Bilateral, symmetrical nodules causing hoarseness.

Key Point: Vocal nodules are **bilateral** and **symmetrical**. Asymmetry suggests malignancy.

5. Laryngomalacia

Clinical Focus: The most common congenital abnormality of the larynx.

- **Definition:** Excessive flaccidity of the supraglottic larynx (epiglottis and aryepiglottic folds).
- **Presentation:** Inspiratory stridor in infants. The floppy tissues collapse into the airway during inspiration.
- **Management:** Often resolves spontaneously as the child grows and cartilage stiffens.

6. Laryngoscopy: The Visual Landmarks

Exam Focus: Identify structures in a direct laryngoscopy view.

Visual Checklist (From Superior to Inferior):

1. **Epiglottis:** Leaf-shaped, anterior.
2. **Valleculae:** Depressions on either side of the median glossoepiglottic fold.
3. **Aryepiglottic Folds:** Lateral boundaries of the laryngeal inlet.
4. **False Vocal Cords (Vestibular Folds):** Red, thick, widely apart.
5. **Sinus of Larynx:** Space between false and true cords.
6. **True Vocal Cords (Vocal Folds):** Pearly white, thin, medial to false cords.
7. **Rima Glottidis:** The opening between the true vocal cords.

Warning: Do not confuse False Cords with True Cords.

- **False Cords:** Vestibular folds. Not involved in phonation. Protect the true cords.
- **True Cords:** Vocal folds. Responsible for phonation. Pearly white appearance due to lack of blood vessels in the ligament.

7. Reinke's Space

Clinical Focus: The potential space under the epithelium of the vocal cord.

- **Significance:** Site of edema in **Allergic reactions** (Angioedema).
 - **Mechanism:** Swelling in Reinke's space narrows the glottis rapidly, causing acute airway obstruction.
 - **Exam Link:** Why do allergies cause sudden choking? Because Reinke's space fills with fluid, swelling the vocal cords inward.
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Summary of High-Yield Exam Points

Exam Focus:

- **Primary Function:** Protection of lower respiratory tract.
- **Narrowest Part:** Rima Glottidis.
- **Only Abductor:** Posterior Cricothyroid muscle (supplied by RLN).
- **Only Tensor (High Pitch):** Cricothyroid muscle (supplied by External Laryngeal Nerve).
- **Cancer Site:** Glottic (SCC), good prognosis due to poor lymphatics.
- **Nodules Site:** Junction of anterior 1/3 and posterior 2/3 of true cords.
- **Bilateral RLN Injury:** Asphyxia (paramedian paralysis).
- **Unilateral RLN Injury:** Hoarseness (paramedian paralysis).

Source Note: All clinical correlations are derived from standard anatomical relationships and physiological mechanisms described in the source text, including nerve supply, lymphatic drainage patterns, and structural boundaries.

One-Page Quick Revision

High-Yield Snapshot

- **Functions:** Protection (Primary), Phonation, Respiration, Deglutition.
- **Location:** Anterior neck, C3–C6 vertebrae.
- **Skeleton:** 9 cartilages (3 unpaired, 3 paired). Hyoid is associated but not part of the larynx.
- **Narrowest Point:** Rima glottidis (critical for airway obstruction).
- **Primary Function:** Protection of lower respiratory tract (sphincter action).

Cartilages & Types

- **Unpaired:**
 - Thyroid: Shield, Adam's apple (prominent in males, 90° angle).
 - Cricoid: Ring, "Foundation stone," encircles lumen completely.
 - Epiglottis: Leaf-like, elastic cartilage, prevents aspiration.
- **Paired:**
 - Arytenoid: Pyramidal, articulates with cricoid.
 - Corniculate: Articulates with arytenoid apex.
 - Cuneiform: Rod-shaped, in aryepiglottic fold.
- **Histology:**
 - Hyaline: Thyroid, Cricoid, Arytenoid base (ossifies >25 yrs).
 - Elastic: Epiglottis, Arytenoid apex, Corniculate, Cuneiform (does not ossify).

Joints & Movements

- **Cricothyroid:** Rotatory/Gliding. RLN enters near here.
- **Cricoarytenoid:** Rotatory/Gliding. Abducts/Adducts cords.
- **Arytenocorniculate:** Non-functional synovial joint.

Membranes & Ligaments

- **Intrinsic:**
 - Quadrangular membrane: Forms vestibular (false) fold.
 - Cricovocal membrane (Conus Elasticus): Forms vocal (true) fold.
- **Extrinsic:**
 - Thyrohyoid: Pierced by Internal Laryngeal Nerve (ILN) + Superior Laryngeal Vessels.
 - Cricothyroid: Pierced by External Laryngeal Nerve (ELN) + Inferior Laryngeal Vessels.

Muscles of Larynx

- **Abductors:** Posterior Cricoarytenoid (only abductor).
- **Adductors:** Lateral Cricoarytenoid, Transverse Arytenoid, Interarytenoid.
- **Tensors:** Cricothyroid (tenses vocal ligament), Vocalis (modulates/tenses).
- **Relaxers:** Thyroarytenoid (relaxes vocal ligament).
- **Inlet Control:** Oblique Arytenoid, Aryepiglotticus.

Innervation

- **Internal Laryngeal Nerve (ILN):** Sensory above vocal cords. Pierces thyrohyoid membrane.
- **External Laryngeal Nerve (ELN):** Motor to Cricothyroid muscle.
- **Recurrent Laryngeal Nerve (RLN):** Motor to all intrinsic muscles except CT. Sensory below vocal cords.

Clinical Correlations

- **Heimlich Maneuver:** Dislodges foreign body from laryngeal inlet.
- **Vocal Nodules:** "Singer's nodules" at junction of anterior 1/3 and posterior 2/3 of vocal cords.
- **Cancer:** Squamous cell carcinoma (90–95%). Glottic region is most common site.
- **Nerve Injury:**
 - Unilateral RLN: Hoarseness.
 - Bilateral RLN: Airway obstruction (cords stuck midline).
 - ELN: Loss of voice pitch control (CT muscle paralysis).

Golden Facts to Remember

- **Primary Function:** Protection of the lower respiratory tract (not phonation).
- **Foundation Stone:** Cricoid cartilage.
- **Only Abductor:** Posterior cricoarytenoid muscle.

- **Only Unpaired Adductor:** Transverse arytenoid muscle.
 - **Tuning Fork:** Cricothyroid muscle (tenses vocal cords for high pitch).
 - **Narrowest Part:** Rima glottidis.
 - **Safety Muscles:** Posterior cricoarytenoids (abduct cords to open airway).
 - **Nerve Supply Exception:** Cricothyroid is supplied by External Laryngeal Nerve; all other intrinsic muscles by Recurrent Laryngeal Nerve.
 - **Ossification:** Hyaline cartilages (Thyroid, Cricoid) ossify after age 25; Elastic cartilages do not.
 - **Male vs. Female Angle:** Thyroid angle is 90° in males (acute) and 120° in females (obtuse).
 - **Vocal Nodule Site:** Junction of anterior one-third and posterior two-thirds of vocal cords.
 - **Most Common Cancer:** Squamous cell carcinoma (SCC).
 - **Most Common Site for Cancer:** Glottic region.
 - **Most Common Congenital Abnormality:** Laryngomalacia.
 - **Reinke's Space:** Potential space under the epithelium of the vocal cord.
 - **Adam's Apple:** Prominence of the thyroid cartilage angle in males.
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